Notebook

1 Introduction

2 Screened longitudinal friction kernel

2.1 Kernel and Gaussian-scale representation

We consider the purely longitudinal matrix-valued kernel

$$K_\kappa(\boldsymbol{r}) = \frac{e^{-\kappa r}}{r} P(\boldsymbol{r}), \qquad P(\boldsymbol{r}) = \frac{\boldsymbol{r}\boldsymbol{r}^\mathsf{T}}{r^2}, \qquad r = |\boldsymbol{r}|, \qquad \kappa > 0. \tag{1}$$

Throughout, $\boldsymbol{k}$ denotes the Fourier wave vector, $k = |\boldsymbol{k}| \geq 0$, and $\xi > 0$ is the Ewald splitting parameter.

Proposition 2.1 (Gaussian-scale representation)**.** *The kernel in Eq. (1) satisfies*

$$K_\kappa(\boldsymbol{r}) = \boldsymbol{r}\boldsymbol{r}^\mathsf{T} \int_0^\infty \rho_\kappa(s) e^{-sr^2}\, ds, \tag{2}$$

with the nonnegative weight

$$\rho_\kappa(s) = \frac{1}{\sqrt{\pi}} \int_0^s t^{-1/2} e^{-\kappa^2/(4t)}\, dt = \frac{2\sqrt{s}}{\sqrt{\pi}} e^{-\kappa^2/(4s)} + \kappa \left[\operatorname{erf}\left(\frac{\kappa}{2\sqrt{s}} \right) - 1 \right]. \tag{3}$$

The proof is given in Appendix A.

2.2 Ewald splitting

We split the Gaussian scales in Eq. (2) at $s = \xi^2$:

$$K_\kappa^L(\boldsymbol{r}) = \boldsymbol{r}\boldsymbol{r}^\mathsf{T} \int_0^{\xi^2} \rho_\kappa(s) e^{-sr^2}\, ds, \tag{4}$$

$$K_\kappa^S(\boldsymbol{r}) = \boldsymbol{r}\boldsymbol{r}^\mathsf{T} \int_{\xi^2}^\infty \rho_\kappa(s) e^{-sr^2}\, ds, \tag{5}$$

o that $K_\kappa = K_\kappa^S + K_\kappa^L$. Since $\boldsymbol{r}\boldsymbol{r}^\mathsf{T} = r^2 P(\boldsymbol{r})$, both pieces remain purely longitudinal,

$$K_\kappa^{S/L}(\boldsymbol{r}) = \theta_{S/L}(r) P(\boldsymbol{r}), \tag{6}$$

where

$$\theta_L(r) = r^2 \int_0^{\xi^2} \rho_\kappa(s) e^{-sr^2}\, ds, \tag{7}$$

$$\theta_S(r) = r^2 \int_{\xi^2}^\infty \rho_\kappa(s) e^{-sr^2}\, ds. \tag{8}$$

Because $\rho_\kappa(s) \geq 0$, both $\theta_L(r) \geq 0$ and $\theta_S(r) \geq 0$, so each real-space piece is pointwise positive semidefinite. Closed-form expressions for θ_L and θ_S are given in Appendix B.

2.3 Localization in real and Fourier space

We use the Fourier convention

$$\widehat{f}(\boldsymbol{k}) = \int_{\mathbb{R}^3} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}} f(\boldsymbol{r})\, d\boldsymbol{r}. \tag{9}$$

Proposition 2.2 (Gaussian localization)**.** *The short-range kernel satisfies*

$$K_\kappa^S(\boldsymbol{r}) \sim \rho_\kappa(\xi^2) e^{-\xi^2 r^2} P(\boldsymbol{r}), \qquad r \to \infty. \tag{10}$$

The long-range kernel has the Fourier transform

$$\widehat{K}_\kappa^L(\boldsymbol{k}) = A_{\kappa,\xi}(k) I + B_{\kappa,\xi}(k) \boldsymbol{k}\boldsymbol{k}^{\mathsf{T}}, \tag{11}$$

where

$$A_{\kappa,\xi}(k) = \frac{\pi^{3/2}}{2} \int_0^{\xi^2} \rho_\kappa(s) s^{-5/2} e^{-k^2/(4s)}\, ds, \tag{12}$$

$$B_{\kappa,\xi}(k) = -\frac{\pi^{3/2}}{4} \int_0^{\xi^2} \rho_\kappa(s) s^{-7/2} e^{-k^2/(4s)}\, ds, \tag{13}$$

and $B_{\kappa,\xi}(k) = A'_{\kappa,\xi}(k)/k$ *for* $k > 0$*. As* $k \to \infty$*,*

$$A_{\kappa,\xi}(k) \sim \frac{2\pi^{3/2}}{\xi k^2} \rho_\kappa(\xi^2) e^{-k^2/(4\xi^2)}, \qquad A_{\kappa,\xi}(k) + k^2 B_{\kappa,\xi}(k) \sim -\frac{\pi^{3/2}}{\xi^3} \rho_\kappa(\xi^2) e^{-k^2/(4\xi^2)}, \tag{14}$$

and consequently

$$\left\| \widehat{K}_\kappa^L(\boldsymbol{k}) \right\|_2 \sim \frac{\pi^{3/2}}{\xi^3} \rho_\kappa(\xi^2) e^{-k^2/(4\xi^2)}, \qquad k \to \infty. \tag{15}$$

The proof is given in Appendix C. The value of $\widehat{K}_\kappa^L$ at $k = 0$ is obtained directly from the integral formulas.

The truncation errors of the two pieces are measured by the continuum tail quantities

$$E_S(r_c) = \int_{|\boldsymbol{r}|>r_c} \|K_\kappa^S(\boldsymbol{r})\|_2\, d\boldsymbol{r}, \qquad E_F(k_c) = \frac{1}{(2\pi)^3} \int_{|\boldsymbol{k}|>k_c} \left\| \widehat{K}_\kappa^L(\boldsymbol{k}) \right\|_2 d\boldsymbol{k}. \tag{16}$$

By Proposition 2.2 (see Appendix C),

$$E_S(r_c) \sim \frac{2\pi}{\xi^2} \rho_\kappa(\xi^2) r_c e^{-\xi^2 r_c^2}, \qquad E_F(k_c) \sim \frac{1}{\sqrt{\pi}\,\xi} \rho_\kappa(\xi^2) k_c e^{-k_c^2/(4\xi^2)}. \tag{17}$$

The quantity E_F is only a continuum asymptotic proxy for the reciprocal tail. A fully discrete periodic PPPM error estimate also depends on the reciprocal lattice, mesh spacing, assignment order, aliasing, and interpolation; these effects are analyzed in Section 3.

2.4 Leading-order balance

Balancing the dominant exponential rates $e^{-\xi^2 r_c^2}$ and $e^{-k_c^2/(4\xi^2)}$ in Eq. (17) gives

$$\xi r_c \approx \frac{k_c}{2\xi}, \qquad \text{or equivalently} \qquad \xi^2 \approx \frac{k_c}{2r_c}. \tag{18}$$

Increasing ξ makes the real-space part more localized but broadens the reciprocal-space contribution, while decreasing ξ does the opposite.

2.5 Pointwise positivity versus Fourier-multiplier signs

The real-space positivity of the split kernels should not be confused with positive semidefiniteness of the Fourier multiplier. By Eq. (11), the eigenvalue of $\widehat{K}_\kappa^L(\boldsymbol{k})$ on vectors orthogonal to $\boldsymbol{k}$ is $\lambda_\perp(k) = A_{\kappa,\xi}(k)$, whereas the eigenvalue in the $\boldsymbol{k}$ direction is

$$\lambda_\parallel(k) = A_{\kappa,\xi}(k) + k^2 B_{\kappa,\xi}(k) = A_{\kappa,\xi}(k) + kA'_{\kappa,\xi}(k). \tag{19}$$

By Eq. (14), $\lambda_\parallel(k) < 0$ for large k, so the Fourier multiplier is generally not positive semidefinite. This does not contradict the pointwise real-space property $K_\kappa^L(\boldsymbol{r}) \succeq 0$. Only the latter enters the graph-Laplacian argument below.

3 Fast periodic graph-Laplacian action

This section concerns the particle graph and its particle–mesh discretization. We use only two properties established in Section 2: the periodic real-space long-range pair kernel $K_\kappa^{L,\mathrm{per}}$ is pointwise positive semidefinite, and its Fourier multiplier has the Gaussian decay of Eq. (15).

3.1 Finite reciprocal truncation and a spectral-gap criterion

For a cubic periodic cell of volume $V = L^3$, let

$$\mathcal{K} = \frac{2\pi}{L}\mathbb{Z}^3. \tag{20}$$

The exact periodic long-range kernel is represented by

$$K_\kappa^{L,\mathrm{per}}(\boldsymbol{r}) = \frac{1}{V}\sum_{\boldsymbol{k}\in\mathcal{K}} \widehat{K}_\kappa^L(\boldsymbol{k}) e^{i\boldsymbol{k}\cdot\boldsymbol{r}}. \tag{21}$$

After a reciprocal cutoff k_c,

$$K_\kappa^{L,k_c}(\boldsymbol{r}) = \frac{1}{V}\sum_{|\boldsymbol{k}|\le k_c} \widehat{K}_\kappa^L(\boldsymbol{k}) e^{i\boldsymbol{k}\cdot\boldsymbol{r}}. \tag{22}$$

Define the omitted tail

$$R_{k_c}(\boldsymbol{r}) = K_\kappa^{L,k_c}(\boldsymbol{r}) - K_\kappa^{L,\mathrm{per}}(\boldsymbol{r}). \tag{23}$$

Then

$$\|R_{k_c}(\boldsymbol{r})\|_2 \le \frac{1}{V}\sum_{|\boldsymbol{k}|>k_c} \|\widehat{K}_\kappa^L(\boldsymbol{k})\|_2. \tag{24}$$

If $\Gamma_L^{k_c}$ is formed from the truncated pair response using the same degree-minus-weight construction, Lemma D.1 in Appendix D gives

$$\|\Gamma_L^{k_c} - \Gamma_L\|_2 \le 2\max_i \sum_{j\neq i} \|R_{k_c}(q_i - q_j)\|_2. \tag{25}$$

A configuration-independent but generally pessimistic consequence is

$$\|\Gamma_L^{k_c} - \Gamma_L\|_2 \le \frac{2(N-1)}{V}\sum_{|\boldsymbol{k}|>k_c} \|\widehat{K}_\kappa^L(\boldsymbol{k})\|_2. \tag{26}$$

The Gaussian Fourier asymptotics imply the expected fixed-system rate

$$\|\Gamma_L^{k_c} - \Gamma_L\|_2 = \mathcal{O}\Big(k_c e^{-k_c^2/(4\xi^2)}\Big), \tag{27}$$

although the worst-case prefactor inherited from Eq. (26) can be very conservative.

Let

$$\mathcal{T} = \mathrm{span}\{\mathbf{1} \otimes e_1, \mathbf{1} \otimes e_2, \mathbf{1} \otimes e_3\}. \tag{28}$$

Assume for the moment that the exact graph has no null vectors beyond translations, and define the restricted spectral gap

$$\lambda_* = \min_{\substack{X \perp \mathcal{T} \\ \|X\|=1}} X^\mathsf{T} \Gamma_L X > 0. \tag{29}$$

Proposition 3.1 (PSD preservation under a small reciprocal perturbation)**.** *If* $\Gamma_L^{k_c}$ *preserves the translation nullspace and*

$$\|\Gamma_L^{k_c} - \Gamma_L\|_2 < \lambda_*, \tag{30}$$

then $\Gamma_L^{k_c} \succeq 0$.

Proof. For $X \perp \mathcal{T}$ with $\|X\| = 1$,

$$X^\mathsf{T} \Gamma_L^{k_c} X = X^\mathsf{T} \Gamma_L X + X^\mathsf{T} (\Gamma_L^{k_c} - \Gamma_L) X \tag{31}$$

$$\geq \lambda_* - \|\Gamma_L^{k_c} - \Gamma_L\|_2 > 0. \tag{32}$$

On $\mathcal{T}$, both operators vanish. □

The condition in Proposition 3.1 is sufficient but not necessary. A truncated pair kernel can fail to be pointwise semidefinite while the corresponding graph Laplacian remains positive semidefinite.

3.2 Symmetric particle–mesh discretization

Let P_h denote spreading from particles to a periodic mesh and let

$$I_h = P_h^\mathsf{T} \tag{33}$$

be the adjoint gathering operator with consistent normalization. Let F_h be the discrete Fourier transform and let $G_h(\boldsymbol{k})$ be an even, real-symmetric tensor influence function. The particle response is

$$W_h = I_h F_h^{-1} G_h F_h P_h. \tag{34}$$

Adjoint transfer and an even symmetric multiplier imply that W_h is symmetric. Rather than using W_h directly as a friction matrix, define

$$(\widetilde{\Gamma}_h^L X)_i = \left(\sum_j (W_h)_{ij} \right) x_i - \sum_j (W_h)_{ij} x_j. \tag{35}$$

The self response cancels algebraically, so the block-row sums vanish exactly and the translation modes are preserved for every mesh and particle configuration. Symmetry and row-zero structure, however, do not by themselves guarantee positive semidefiniteness.

If K_{ij}^h denotes the complete approximate pair response and

$$E_{ij} = K_{ij}^h - K_{ij}, \tag{36}$$

define

$$\epsilon_{\mathrm{op}} = 2 \max_i \sum_{j \neq i} \|E_{ij}\|_2. \tag{37}$$

Lemma D.1 gives

$$\|\widetilde{\Gamma}_h - \Gamma\|_2 \leq \epsilon_{\mathrm{op}}. \tag{38}$$

Therefore, if the exact graph has restricted gap λ_*, defined in Eq. (29), and

$$\epsilon_{\mathrm{op}} < \lambda_*, \tag{39}$$

then the raw particle–mesh graph is already positive semidefinite and no stabilization is required.

If this sufficient certificate is not available, one may retain the translation-preserving fallback stabilization

$$\Gamma_h = \widetilde{\Gamma}_h + \delta_h \Pi, \qquad \Pi = \left(I_N - \frac{1}{N} \mathbf{1}\mathbf{1}^{\mathsf{T}} \right) \otimes I_3, \tag{40}$$

with

$$\delta_h \geq \max \left\{ 0, -\lambda_{\min} \left(\widetilde{\Gamma}_h |_{\mathrm{Range}\, \Pi} \right) \right\}. \tag{41}$$

Proposition 3.2 (Stabilized operator)**.** *The operator Γ_h is symmetric positive semidefinite, satisfies $\Gamma_h t = 0$ for every $t \in \mathcal{T}$, and obeys*

$$\|\Gamma_h - \Gamma\|_2 \leq \epsilon_{\mathrm{op}} + \delta_h. \tag{42}$$

The proof is given in Appendix D. The conservative choice $\delta_h = \epsilon_{\mathrm{op}}$ is always certified, but the numerical tests in Section 6.1 show that it is usually much more conservative than the actual spectral shift.

3.3 Complete off-grid mesh-response decomposition

The mesh error is most transparent when the aliases are kept explicitly. Let

$$\boldsymbol{k}_n = \boldsymbol{k} + \frac{2\pi}{h} n, \qquad n \in \mathbb{Z}^3, \tag{43}$$

and let $\widehat{W}_h$ be the Fourier transform of the assignment window. For a particle at q, define the deposited amplitude

$$S_h(\boldsymbol{k}, q) = \sum_{n \in \mathbb{Z}^3} \widehat{W}_h(\boldsymbol{k}_n) e^{-i \boldsymbol{k}_n \cdot q}. \tag{44}$$

Separate the principal contribution and the aliases:

$$S_0(\boldsymbol{k}, q) = \widehat{W}_h(\boldsymbol{k}) e^{-i \boldsymbol{k} \cdot q}, \tag{45}$$

$$A_h(\boldsymbol{k}, q) = \sum_{n \neq 0} \widehat{W}_h(\boldsymbol{k}_n) e^{-i \boldsymbol{k}_n \cdot q}. \tag{46}$$

Define the relative alias factor

$$\alpha_h(\boldsymbol{k}, q) = \frac{A_h(\boldsymbol{k}, q)}{\widehat{W}_h(\boldsymbol{k}) e^{-i \boldsymbol{k} \cdot q}}, \tag{47}$$

so that

$$S_h(\boldsymbol{k}, q) = \widehat{W}_h(\boldsymbol{k}) e^{-i \boldsymbol{k} \cdot q} \left[1 + \alpha_h(\boldsymbol{k}, q) \right]. \tag{48}$$

The phase factor

$$e^{-i(2\pi/h)n\cdot q} \tag{49}$$

inside α_h shows explicitly that a finite mesh response depends on particle positions relative to the grid, not only on pair separation.

Define

$$H_h(\boldsymbol{k}) = |\widehat{W}_h(\boldsymbol{k})|^2 G_h(\boldsymbol{k}), \qquad D_h(\boldsymbol{k}) = H_h(\boldsymbol{k}) - \widehat{K}_\kappa^L(\boldsymbol{k}). \tag{50}$$

A direct computation (Appendix E) gives the exact single-mode pair-response error

$$E_{ij}^h(\boldsymbol{k}) = e^{i\boldsymbol{k}\cdot(q_i-q_j)}\left[D_h(\boldsymbol{k}) + H_h(\boldsymbol{k})\left(\alpha_i^* + \alpha_j + \alpha_i^*\alpha_j\right)\right], \tag{51}$$

where $\alpha_i = \alpha_h(\boldsymbol{k}, q_i)$ and similarly for α_j. Equation (51) separates an alias-free influence defect D_h from the complete off-grid alias response.

For ideal deconvolution,

$$G_h(\boldsymbol{k}) = \frac{\widehat{K}_\kappa^L(\boldsymbol{k})}{|\widehat{W}_h(\boldsymbol{k})|^2}, \tag{52}$$

whenever the denominator is numerically safe. Then

$$D_h(\boldsymbol{k}) = 0 \tag{53}$$

in exact arithmetic, and the complete mesh error is produced entirely by the alias cross terms

$$\widehat{K}_\kappa^L(\boldsymbol{k})\left(\alpha_i^* + \alpha_j + \alpha_i^*\alpha_j\right). \tag{54}$$

The linear terms are first order in the alias amplitude, while the quadratic term is second order. This distinction is important when defining numerical diagnostics: a diagnostic retaining only matched quadratic aliases can scale like h^{2p} and miss the dominant h^p contribution.

3.4 B-spline assignment and the h^p mesh law

For a cardinal B-spline assignment of order p,

$$\widehat{W}_h(\boldsymbol{k}) = \prod_{\ell=1}^{3} \operatorname{sinc}^p\left(\frac{k_\ell h}{2}\right), \qquad \operatorname{sinc}(x) = \frac{\sin x}{x}. \tag{55}$$

For $p \geq 2$, the relative alias amplitude satisfies

$$\alpha_h(\boldsymbol{k}, q) = \mathcal{O}((h|\boldsymbol{k}|)^p). \tag{56}$$

Because the screened long-range Fourier multiplier has Gaussian decay, all moments

$$M_p(\kappa, \xi) = \frac{1}{(2\pi)^3}\int_{\mathbb{R}^3} k^p \|\widehat{K}_\kappa^L(\boldsymbol{k})\|_2 \, d\boldsymbol{k} \tag{57}$$

are finite. Therefore the complete off-grid mesh pair response obeys the asymptotic law

$$\epsilon_{\text{mesh}} = \mathcal{O}(h^p) \tag{58}$$

for fixed physical cutoff and sufficiently fine mesh; the proof is given in Appendix E.

The phase dependence in α_h implies oscillations in a single fixed configuration as h varies. Random global grid offsets remove the mean of the linear phase terms but not their RMS magnitude. Consequently, RMS phase-averaged errors give a cleaner estimate of the asymptotic h^p behavior.

It is useful to write the measured asymptotic law in dimensionless form as

$$\epsilon_{\mathrm{mesh}} \approx C_M(\kappa, \xi, p, k_c, \text{configuration})(\xi h)^p. \tag{59}$$

The coefficient in Eq. (59) is not universal; the experiments in Section 6.1 show a modest dependence on configuration and reciprocal cutoff. Parameter selection should therefore use a conservative envelope $C_M^{\max}(p)$ rather than a single universal constant.

4 Discrete FDT thermostat and time integration

4.1 A finite-time thermostat map

The matrix-free Ewald–PPPM formulation of Section 3 provides a fast action of the discrete friction operator,

$$\boldsymbol{x} \mapsto \Gamma_h \boldsymbol{x}, \tag{60}$$

with measured cost consistent with

$$T_{\Gamma_h}(N) = \mathcal{O}(N \log N). \tag{61}$$

The remaining difficulty is the correlated thermal noise. For the exponential finite-time thermostat, the damping and noise factors are

$$R_{\Delta t}(\Gamma_h) = \exp\left(-\frac{\Delta t}{m}\Gamma_h\right), \tag{62}$$

and

$$S_{\Delta t}(\Gamma_h) = \left[\beta^{-1} m \left(I - R_{\Delta t}(\Gamma_h)^2\right)\right]^{1/2}. \tag{63}$$

The pair $(R_{\Delta t}, S_{\Delta t})$ reproduces the exact transition law of the frozen-configuration Ornstein–Uhlenbeck flow, conserves total momentum pathwise, and leaves the conditional Maxwell law invariant; see Appendix F. The random increment is therefore

$$\eta = f_{\Delta t}(\Gamma_h)\, \xi, \tag{64}$$

with

$$f_{\Delta t}(\lambda) = \sqrt{\beta^{-1} m \left(1 - e^{-2\Delta t \lambda/m}\right)}. \tag{65}$$

We evaluate this matrix-function action by the Lanczos method. If the required Krylov rank is $r(N)$, the noise cost is

$$T_{\mathrm{noise}}(N) = \mathcal{O}(r(N) N \log N)\,. \tag{66}$$

Consequently, a measured $\mathcal{O}(N \log N)$ production cost requires that the rank remain effectively constant under a fixed accuracy budget.

4.2 Lanczos evaluation of the correlated noise

We first consider direct-start Lanczos with

$$z = \Pi\xi, \qquad \Pi = \left(I - \frac{1}{N}\mathbf{1}\mathbf{1}^{\mathsf{T}}\right) \otimes I_3, \tag{67}$$

and

$$q_1 = \frac{z}{\|z\|}, \qquad \eta_r = \|z\| Q_r f_{\Delta t}(T_r) e_1. \tag{68}$$

The projection Π is applied after every Γ_h action and after the final reconstruction.

4.3 Range-start Lanczos

A Landau-style range start is a natural alternative. Defining

$$b = \Gamma_h z, \tag{69}$$

and

$$g(\lambda) = \frac{f_{\Delta t}(\lambda)}{\lambda}, \tag{70}$$

one may write, on the positive spectral range,

$$f_{\Delta t}(\Gamma_h) z = g(\Gamma_h) b. \tag{71}$$

Although $b = \Gamma_h z$ strongly suppresses the smallest spectral modes in the starting vector, the transformed scalar function behaves as

$$g(\lambda) \sim C\lambda^{-1/2}, \qquad \lambda \to 0^+, \tag{72}$$

because

$$f_{\Delta t}(\lambda) \sim C\sqrt{\lambda}, \tag{73}$$

with $C = (2\beta^{-1}\Delta t)^{1/2}$ (Appendix F). The increased singularity of g more than offsets the spectral reweighting.

4.4 Full symmetric splitting

The PPPM–Lanczos thermostat is inserted into the symmetric Langevin splitting

$$\Psi_{\Delta t} = B_{\Delta t/2} A_{\Delta t/2} O_{\Delta t} A_{\Delta t/2} B_{\Delta t/2}. \tag{74}$$

For the ideal gas, $U = 0$ and the conservative B steps vanish. One step reduces to

$$q^{(1)} = q^n + \frac{\Delta t}{2m} p^n, \tag{75}$$

$$p^{n+1} = e^{-\Delta t \Gamma_h(q^{(1)})/m} p^n + f_{\Delta t}(\Gamma_h(q^{(1)}))\xi^n, \tag{76}$$

and

$$q^{n+1} = q^{(1)} + \frac{\Delta t}{2m} p^{n+1}, \tag{77}$$

with periodic wrapping of the positions.

Both the deterministic damping action and the stochastic noise action are evaluated matrix-free. The center-of-mass momentum component is copied exactly through the damping step, while the fluctuating component and every noise sample are projected onto Range(Π).

5 Simulation error and complexity

5.1 Operator error budget

For the present ideal-deconvolution implementation, it is more natural to combine aliasing and off-grid transfer into one complete mesh-response term:

$$\epsilon_{\text{op}} \le \epsilon_S + \epsilon_F + \epsilon_{\text{mesh}} + \epsilon_{\text{quad}}. \tag{78}$$

For the screened longitudinal kernel considered here the Fourier multiplier is available analytically, so ϵ_{quad} can be made negligible. With the balanced variable

$$s = \xi r_c = \frac{k_c}{2\xi}, \tag{79}$$

the real- and reciprocal-space Ewald tails have the common Gaussian rate

$$\epsilon_S + \epsilon_F \sim C_E e^{-s^2} \tag{80}$$

up to algebraic prefactors, whereas for fixed B-spline order

$$\epsilon_{\text{mesh}} \approx C_M (\xi h)^p. \tag{81}$$

If the mesh is parameterized by

$$k_c = \eta_N \frac{\pi}{h}, \qquad 0 < \eta_N < 1, \tag{82}$$

then

$$\xi h = \frac{\eta_N \pi}{2s}, \tag{83}$$

and the schematic total error becomes

$$\epsilon_{\text{op}} \lesssim C_E e^{-s^2} + C_M \left(\frac{\eta_N \pi}{2s}\right)^p. \tag{84}$$

Thus, in Eq. (84), the Ewald tail is Gaussian in s, while the mesh error is only algebraic in s at fixed assignment order. At sufficiently high target accuracy the mesh term must therefore dominate unless p is increased or h is reduced more aggressively.

5.2 Practical parameter selection

A convenient initial Ewald choice is

$$s = \sqrt{\log\left(\frac{2}{\text{tol}}\right)}. \tag{85}$$

The balanced cutoffs are then

$$r_c = \frac{s}{\xi}, \qquad k_c = 2\xi s. \tag{86}$$

The factor ξ should be treated as a computational work-balancing parameter rather than an accuracy parameter. For a chosen B-spline order p, a practical mesh requirement is

$$C_M^{\max}(p)(\xi h)^p \le \frac{\text{tol}}{2}, \tag{87}$$

which gives

$$h \le \frac{1}{\xi}\left(\frac{\text{tol}}{2C_M^{\max}(p)}\right)^{1/p}. \tag{88}$$

This should be combined with a Nyquist safety constraint such as

$$\frac{k_c h}{\pi} \le 0.8. \tag{89}$$

For a box of side length L, one can then choose

$$M \geq \frac{L}{h} \tag{90}$$

and round upward to an FFT-friendly size. This last rounding decreases the actual mesh spacing and therefore does not weaken the accuracy condition. In equal-accuracy timings at $\kappa = 1$ (Appendix G), the accepted Ewald scale was 1.0–1.075 times the initial choice above, and the cost per action was lowest near $\xi = 0.5$–0.7; that accuracy was measured on two reference configurations only.

Once a complete pair-response estimate is available, accuracy is certified through

$$\|\widetilde{\Gamma}_h - \Gamma\|_2 \leq \epsilon_{\mathrm{op}}. \tag{91}$$

If a reliable restricted gap λ_* is also available, the same error estimate provides the optional PSD certificate

$$\epsilon_{\mathrm{op}} < \lambda_* \quad \Longrightarrow \quad \widetilde{\Gamma}_h \succeq 0. \tag{92}$$

Failure of this sufficient inequality does not imply indefiniteness; it only means that the perturbation certificate is inconclusive.

5.3 Work estimate

With fixed noise and damping ranks r_{noise} and r_{damp}, the dominant cost per thermostat step can be summarized as

$$T_{\mathrm{step}} = (r_{\mathrm{noise}} + r_{\mathrm{damp}})\, T_{\Gamma_h} + T_{\mathrm{setup}} + T_{\mathrm{reorth}} + T_{\mathrm{small\ matrix}} + T_{\mathrm{vector}}. \tag{93}$$

In the earlier setting of Section 6 the ranks were $r_{\mathrm{noise}} = 40$ and $r_{\mathrm{damp}} = 16$, i.e. 56 actions per step. Because the measured Γ_h action is $\mathcal{O}(N \log N)$-like (Section 6.4) and the remaining costs are subdominant over the tested regime, the numerical evidence supports the following accuracy-qualified statement.

Remark 5.1 (Measured complexity)**.** *Under the tested fixed physical-accuracy budget, using the fixed Lanczos ranks $r_{\mathrm{noise}} = 40$ and $r_{\mathrm{damp}} = 16$ of the earlier setting, the measured PPPM–Lanczos thermostat cost is consistent with $\mathcal{O}(N \log N)$ scaling for system sizes up to $N = 16000$.*

Remark 5.1 is deliberately finite-range and accuracy-qualified. It is *not* an asymptotic theorem that the rank remains bounded as $N \to \infty$. In particular, the static rank-40 noise error already reaches approximately 9.1×10^{-4} at $N = 16000$, close to the 10^{-3} production budget; if the ranks must grow with N, so does the cost per step.

6 Numerical experiments

This section tests the components of the method on the screened longitudinal kernel of Eq. (1) with $\kappa = 1$, in a fully periodic cube with the $k = 0$ term retained, and with $\beta = m = 1$. The evidence is organized in four groups: operator accuracy and structural checks (Section 6.1); Lanczos accuracy and the choice of production ranks (Section 6.2); conservation, equilibrium and dynamical accuracy of the ideal-gas thermostat (Section 6.3); and computational cost (Section 6.4). The Lennard-Jones application of Section 7 has its own kernel amplitude, screening length, density, time step and ranks, and none of its parameters are used here.

Status of the reported numbers. The results of Sections 6.2–6.4 were obtained with an earlier version of the code, in which $A_{\kappa,\xi}(k)$ and $B_{\kappa,\xi}(k)$ were evaluated at $|\boldsymbol{k}|$ rounded to twelve decimals and the spatial set used $s = 4.100$ instead of the accepted $s = 4.100151562$. Each is reported with its original setting. A recomputation with the corrected code is in progress; entries marked *[Pending]* will be replaced by its results, and nothing from the recomputation is reported before it is complete and checked.

6.1 Operator accuracy and structural checks

[Pending: Experiment 1 is being rerun: the complete production action $X \mapsto \widetilde{\Gamma}_h X$ on static configurations against the unsplit full-periodic reference, with each error source of Eq. (78) measured alone. Its results will replace or supplement the mesh-decomposition study below.]

Mesh response. The error model of Sections 2 and 3 was tested on two periodic configurations with dense references, $\kappa = \xi = 1$, $k_c \in \{4, 6\}$: case A with $N = 30$, $L = 6$, and case B with $N = 40$, $L = 10$. The literal spread–FFT–influence–gather implementation agreed with its matrix representation to approximately 1.4×10^{-15}, and the pair-response error reconstructed from the D_h and α_h terms of Eq. (51) agreed with $K_h - K_F$ to approximately 1.8×10^{-15} relative to the maximum retained pair response.

For ideal deconvolution the largest observed influence defect was $\|D_h\|/\|\widehat{K}^L_\kappa\| \lesssim 1.8 \times 10^{-16}$, and its graph-operator contribution was at most 4.7×10^{-17}, so the observed off-grid error is carried by the alias cross terms. An earlier alias diagnostic retained only the matched quadratic terms $n = n' \neq 0$, which scale as h^{2p}, and omitted the dominant linear terms, which scale as h^p; this explains discrepancies by factors of approximately 22 to 2×10^5 between that diagnostic and the complete mesh response.

On the finest meshes the RMS phase-averaged mesh errors have fitted slopes between p and $p + 1.5$ (Fig. 1). The excess above p decreases as $h \to 0$; the slopes are consistent with the h^p law of Section 3 for approximately $0.1 \lesssim \xi h \lesssim 0.4$ when $p \leq 6$, with a narrower interval for $p = 7, 8$, but finite-mesh fits do not establish the asymptotic order. The normalized quantity $C_M(h) = \epsilon_{\mathrm{mesh}}/(\xi h)^p$ was nearly constant on the finest meshes, typically within factors 1.04–1.4, and within about a factor 2 for $p \geq 7$ in the larger box. Across the tested cases C_M decreased from roughly 2×10^{-2}–3.5×10^{-2} at $p = 2$ to 1.6×10^{-4}–1.1×10^{-3} at $p = 8$, with up to a factor of about three dependence on configuration and k_c.

The predictor $C_M(\xi h)^p$ was substantially more accurate than the edge-of-band heuristic q^p, $q = \eta_N/(2 - \eta_N)$ (Fig. 2). Over 181 mesh runs the ratio of the $C_M(\xi h)^p$ prediction to the measured error had median approximately 0.93 and range 0.25–2.1, whereas q^p overpredicted by a median factor of approximately 1.9×10^3, with range 37 to 4×10^5. The heuristic uses the retained spectral edge, while the complete error is weighted toward smaller k, where the long-range multiplier is larger and the alias ratio much smaller.

Spectrum and positivity. No negative graph eigenvalue was observed among 13 195 mesh operators or among the 44 accepted equal-accuracy runs of Appendix G. The smallest observed ratio was $\lambda_h/\lambda_* \approx 0.970$ (Fig. 3). The spectral-gap certificate was conservative: the actual gap shift was a median of about 2.4% and at most 7% of $\|\Gamma_h - \Gamma\|_2$. Stabilization therefore remains a valid fallback, but it was not required in the tested regime. Symmetry and the translation null space hold by construction (Section 3); their residuals in the dynamic runs are reported in Sections 6.3 and 6.4.

These tests predate the correction of the $|\boldsymbol{k}|$ rounding, which perturbs the multiplier at the level of 10^{-13} relative to $\|\Gamma\|_2$ and does not affect the mesh errors quoted above.

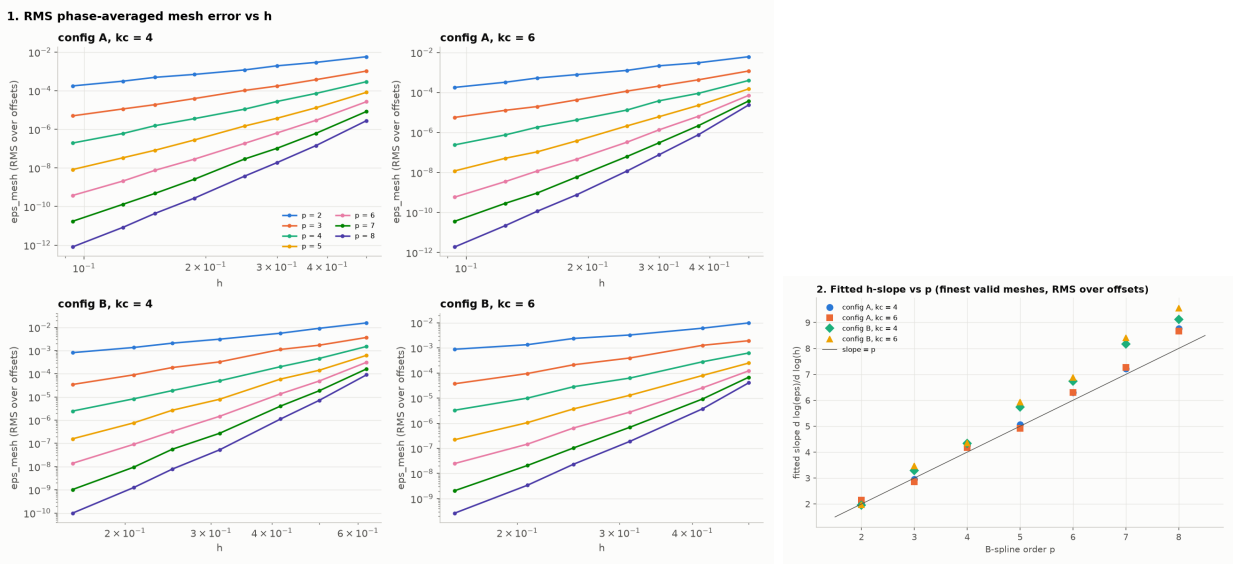


Figure 1: Mesh response, $\kappa = \xi = 1$, $k_c \in \{4, 6\}$, configurations A ($N = 30$, $L = 6$) and B ($N = 40$, $L = 10$). Left: RMS phase-averaged mesh error versus h for $p = 2$–8. Right: fitted slope on the finest valid meshes versus p; the line is slope $= p$. Source: `mesh_error_decomposition_results/plot01, plot02`.

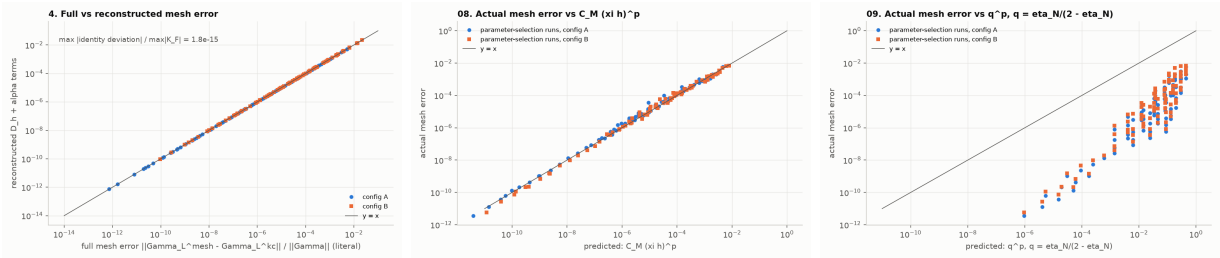


Figure 2: Mesh response, configurations A and B. Left: error reconstructed from the D_h and α_h terms of Eq. (51) versus the literal mesh error. Middle: measured error versus $C_M(\xi h)^p$. Right: measured error versus q^p, $q = \eta_N/(2 - \eta_N)$. Source: `mesh_error_decomposition_results/plot04, plot08, plot09`.

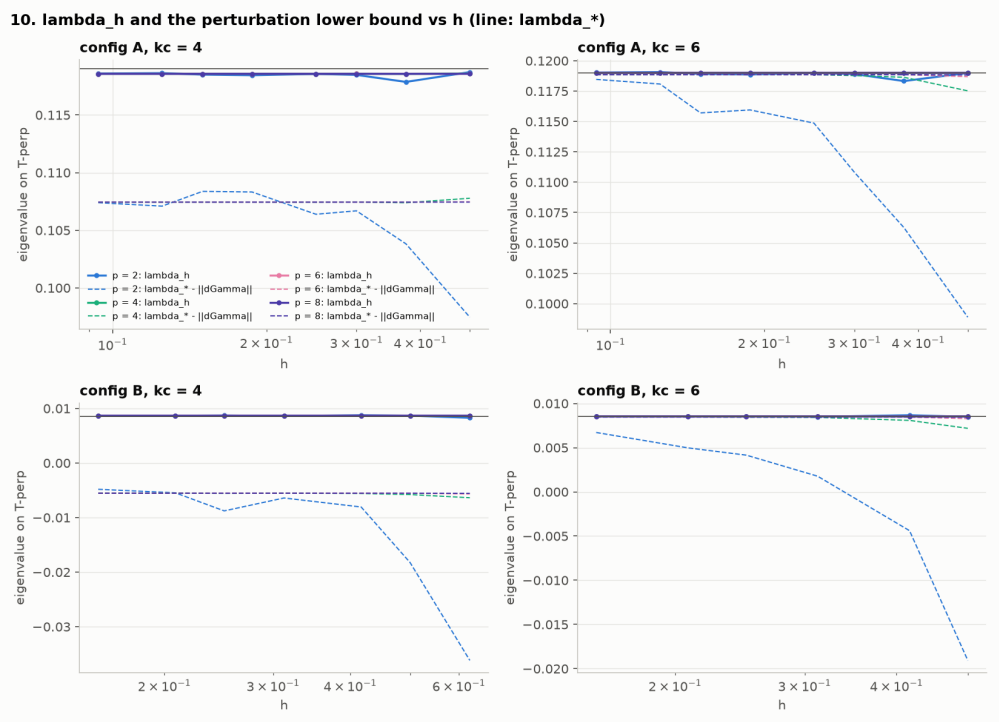


Figure 3: Smallest eigenvalue λ_h of the particle–mesh graph on $\mathcal{T}^{\perp}$ and the perturbation lower bound versus h; the horizontal line is λ_*. Configurations A and B, $k_c \in \{4, 6\}$, several p. Source: `mesh_error_decomposition_results/plot10_lambda_vs_h.png`.

6.2 Lanczos accuracy and production-rank selection

The thermostat applies two matrix functions of the same $\widetilde{\Gamma}_h$: the noise factor $f_{\Delta t}$, which behaves like a square root near $\lambda = 0$, and the damping factor $e^{-\Delta t\widetilde{\Gamma}_h/m}$. Lanczos errors are measured as relative action errors against converged high-rank Lanczos references of the same operator, so they contain no spatial error. A reference was accepted only when both the successive-rank difference and the rank-r versus rank-$(r-20)$ difference satisfied stringent convergence criteria; for $N = 8000$ and 16000 this required references up to rank 360. Two questions are kept apart: how the rank grows at a fixed high accuracy, and whether one fixed rank meets the production budget over the tested range.

Rank growth at high accuracy. At relative noise-action tolerance 10^{-8} the required direct-start rank grows with N (Table 1, Fig. 4), approximately as $r(N) \sim N^{0.34}$. A claim $T_{\text{thermostat}} = \mathcal{O}(N \log N)$ at fixed 10^{-8} matrix-function accuracy is therefore not supported. The range start of Section 4.3 required more steps than the direct start: at $N = 4000$ and 10^{-8}, about 166–169 versus 126–128, and typically 20%–35% more Γ_h actions across the tested sizes. The direct start is retained.

N	128	256	512	1024	2048	4000
r	39	50	64	83	102	127

Table 1: Representative direct-start Lanczos ranks at relative noise-action tolerance 10^{-8} (earlier setting).

Fixed noise rank under the production budget. The production target is a relative noise-action error $\varepsilon_{\text{root}} \le 10^{-3}$. Table 2 lists the largest error over the tested Gaussian seeds at $\Delta t\lambda_{\max}/m = 0.5$ for $N \le 4000$, and Table 3 extends the study to $N = 8000$ and 16000 with three seeds per size. Rank 24 does not meet the budget uniformly; rank 32 meets it through $N = 4000$ but fails at $N = 8000$ and 16000. Rank 40 is the smallest single tested rank that stays below 10^{-3} at every tested size, with a thin margin at $N = 16000$ (largest error 9.1×10^{-4}). This is a finite-range statement at fixed accuracy, not a bound on the rank as $N \to \infty$.

N	$r = 16$	$r = 24$	$r = 32$	$r = 40$
128	1.8×10^{-4}	6.1×10^{-6}	1.5×10^{-7}	4.6×10^{-9}
256	4.8×10^{-4}	4.3×10^{-5}	3.4×10^{-6}	2.6×10^{-7}
512	1.4×10^{-3}	1.8×10^{-4}	2.0×10^{-5}	3.3×10^{-6}
1024	2.2×10^{-3}	4.8×10^{-4}	9.6×10^{-5}	1.8×10^{-5}
2048	3.8×10^{-3}	1.0×10^{-3}	2.6×10^{-4}	7.8×10^{-5}
4000	4.6×10^{-3}	1.4×10^{-3}	5.2×10^{-4}	2.0×10^{-4}

Table 2: Fixed-rank direct-start noise-action errors for $N \le 4000$ (maximum over seeds, earlier setting).

Damping rank. The damping factor is smoother than the noise factor. Against a converged $e^{-\Delta t\widetilde{\Gamma}_h/m}\Pi p$, rank 12 gave relative errors of at most 7×10^{-7} up to $\Delta t\lambda_{\max} = 10$, but 1.8×10^{-4} at

N	$r = 32$	$r = 40$	$r = 48$
128	1.5×10^{-7}	4.6×10^{-9}	$\leq 4.6 \times 10^{-9}$
256	3.4×10^{-6}	2.6×10^{-7}	$\leq 2.6 \times 10^{-7}$
512	2.0×10^{-5}	3.3×10^{-6}	$\leq 3.3 \times 10^{-6}$
1024	9.6×10^{-5}	1.8×10^{-5}	$\leq 1.8 \times 10^{-5}$
2048	2.6×10^{-4}	7.8×10^{-5}	$\leq 7.8 \times 10^{-5}$
4000	5.2×10^{-4}	2.0×10^{-4}	$\leq 2.0 \times 10^{-4}$
8000	1.04×10^{-3}	4.5×10^{-4}	2.2×10^{-4}
16000	1.78×10^{-3}	9.1×10^{-4}	4.8×10^{-4}

Table 3: Extended fixed-rank noise-action errors (earlier setting). For $N \leq 4000$ the rank-48 entries are upper bounds from the monotone decrease with rank; rank 48 was not evaluated there.

$\Delta t \lambda_{\max} = 20$; rank 16 stayed below about 6×10^{-7} at $\Delta t \lambda_{\max} = 20$. Against the damping budget 10^{-6}, rank 12 is therefore adequate but marginal for close-pair spikes of $\lambda_{\max}$, and rank 16 leaves a margin.

Production ranks. In the earlier setting the outcome of this section was $r_{\text{noise}} = 40$ and $r_{\text{damp}} = 16$ for the budgets 10^{-3} (noise) and 10^{-6} (damping) through $N = 16000$. These values belong to that setting only. *[Pending: Recommended ranks for the corrected operator are determined by the rank-selection stage of the recomputation.]*

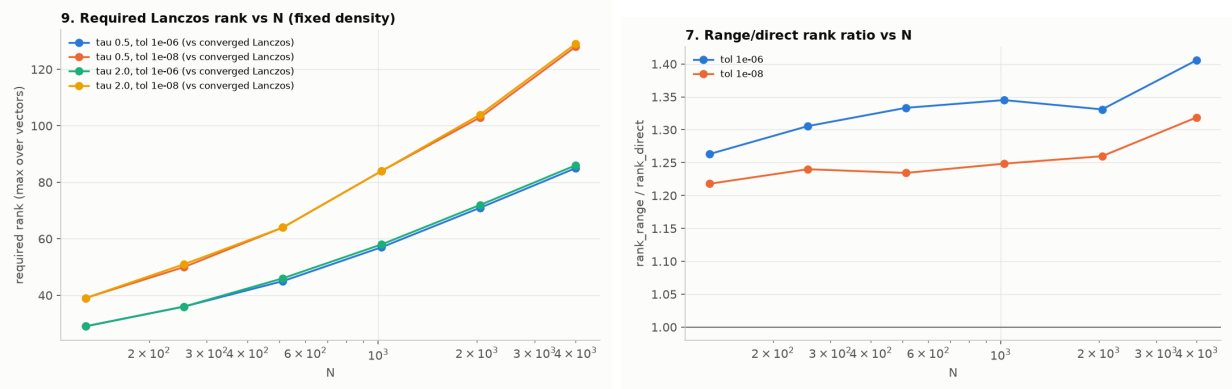


Figure 4: Rank growth at high accuracy. Left: required direct-start rank versus N at fixed density for tolerances 10^{-6} and 10^{-8} and $\Delta t \lambda_{\max}/m = 0.5$, 2.0, against converged references. Right: ratio of range-start to direct-start rank. Source: `lanczos_fdt_results/plot09`, `lanczos_range_start_results/plot07`.

6.3 Conservation, equilibrium, and dynamical accuracy

Setting. The dynamic tests use the ideal gas ($U = 0$) at density 30/216 with the splitting of Section 4.4, the PPPM set $\xi = 0.7$, $s = 4.10$, $\eta_N = 0.7$, $p = 7$, and $\Delta t = 0.0429 = 0.2\, m/\text{median}\, \lambda_{\max}(q_0)$ over the 16 initial configurations at $N = 64$; the same physical Δt is used at every N. The ranks are noise 40 (and 48 for comparison) with damping rank 12; the high-rank Lanczos runs use 90 and 24. These runs therefore used damping rank 12, not the 16 adopted afterwards in Section 6.2 and used for the timings of Section 6.4. At $N = 64$ the coupled methods share initial states and Gaussian

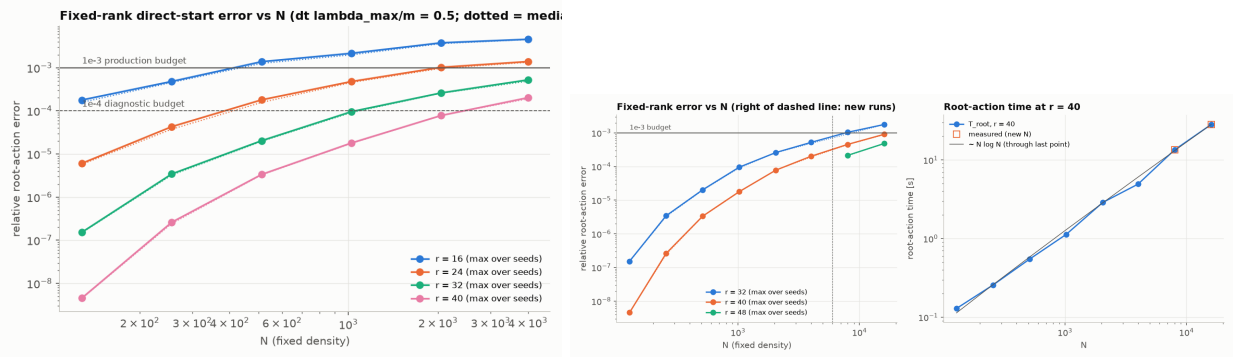


Figure 5: Fixed-rank noise-action error versus N at $\Delta t\lambda_{\max}/m = 0.5$ (maximum over seeds). Left: $N \leq 4000$, ranks 16–40, with the 10^{-3} budget. Right: extension to $N = 8000$ and 16000 (right of the dashed line) for ranks 32, 40, 48, and the rank-40 noise-action time. Source: `fixed_rank_budget_results/`.

streams: dense eigendecomposition of the same $\widetilde{\Gamma}_h$ (the reference), the same with the 10^{-9} spatial set, high-rank Lanczos, ranks 40 and 48, and dense runs at $2\Delta t$, $\Delta t/2$ and $\Delta t/4$, 16 trajectories to $T_{\text{end}} = 6$. Because there is no repulsive potential, close pairs make $\lambda_{\max}$ spike: $\Delta t\lambda_{\max}(t)$ reached about 3.7 at $N = 64$ and 10 at $N = 16000$; the exponential damping step is unconditionally stable and remained so.

Momentum and thermal equilibrium. Total momentum was preserved pathwise to round-off in every run, with largest relative defect 3.8×10^{-14} (Fig. 7, left). A control run with independent, uncorrelated noise drifted by $\mathcal{O}(1)$ (defects 1–1.7). From the deliberately anisotropic start $T_x \approx 1.96$, $T_y \approx T_z \approx 0.5$ the component temperatures relaxed toward $T = 1$ by $t \approx 3$–4 at $N = 64$ (Fig. 6). The window $t \in [3, 6]$ had not yet equilibrated the slowest mode, so 16 longer trajectories to $T = 30$ with burn-in to $t = 15$ were added; in these runs the 95% confidence interval of every component temperature contains 1, and the normalized conditional-Maxwell diagnostics are variance 0.9995 ± 0.010, skewness -0.006 ± 0.007 and excess kurtosis -0.023 ± 0.025 (Fig. 7, right). At $N = 4000$ and 16000 relaxation is visible but equilibrium was not tested.

Separation of error sources. Table 4 lists paired differences of the observables at $N = 64$. The comparisons isolate four contributions:

- Krylov error on the same $\widetilde{\Gamma}_h$: high-rank Lanczos minus dense is at most 1.1×10^{-13}, rank 48 minus dense at most 1.2×10^{-11}, rank 40 minus dense at most 1.0×10^{-9} (Fig. 8);
- spatial discretization: replacing the production set by the 10^{-9} set changes the observables by at most 5.6×10^{-7}. This is a proxy obtained by tightening the set, not a comparison with the exact Γ;
- time discretization: paired differences between dense runs at successive step sizes are about 10^{-2} to 3×10^{-2};
- sampling: the standard error of the mean over 16 trajectories is 3×10^{-2} to 8×10^{-2}.

At larger N, on the same seed and stream, rank 48 minus rank 40 is at most 4.1×10^{-5} at $N = 4000$ and 1.4×10^{-4} at $N = 16000$, compared with seed-to-seed standard deviations of the late-time

temperatures of 0.009–0.019 and 0.002–0.007, respectively. Within the resolution of these observables, rank 40 is indistinguishable from rank 48 through $N = 16000$ and from the dense reference at $N = 64$.

comparison	T_x	T_y	T_z	T_{kin}	O_k
rank 40 − dense	6.6×10^{-10}	6.1×10^{-10}	1.0×10^{-9}	4.1×10^{-10}	5.6×10^{-11}
rank 48 − dense	1.2×10^{-11}	7.3×10^{-12}	5.1×10^{-12}	3.7×10^{-12}	4.3×10^{-13}
high rank − dense	3.0×10^{-14}	1.1×10^{-13}	5.7×10^{-14}	2.3×10^{-14}	7.6×10^{-15}
10^{-9} set − dense	2.4×10^{-7}	3.1×10^{-7}	5.6×10^{-7}	1.7×10^{-7}	4.2×10^{-8}
$\Delta t - \Delta t/2$	2.8×10^{-2}	3.3×10^{-2}	3.4×10^{-2}	1.1×10^{-2}	1.5×10^{-2}
(SE)	2.3×10^{-2}	1.9×10^{-2}	2.4×10^{-2}	1.1×10^{-2}	9×10^{-3}
$\Delta t/2 - \Delta t/4$	2.6×10^{-2}	1.7×10^{-2}	3.5×10^{-2}	1.1×10^{-2}	2.5×10^{-3}
(SE)	1.8×10^{-2}	1.6×10^{-2}	2.4×10^{-2}	1.0×10^{-2}	5×10^{-3}
$2\Delta t - \Delta t$	2.8×10^{-2}	2.7×10^{-2}	4.1×10^{-2}	2.7×10^{-2}	2.9×10^{-3}
(SE)	2.9×10^{-2}	2.6×10^{-2}	3.1×10^{-2}	1.3×10^{-2}	9×10^{-3}
sampling SE of the mean	8.4×10^{-2}	4.9×10^{-2}	6.1×10^{-2}	3.1×10^{-2}	2.9×10^{-2}

Table 4: Paired differences at $N = 64$, 16 trajectories, noise/damping ranks 40/12 and 48/12 (earlier setting). Each entry is the largest over time of the absolute trajectory mean of the paired difference; $O_k = \langle \cos(2\pi x/L) \rangle$. "Dense" is the eigendecomposition of the same $\widetilde{\Gamma}_h$. Source: `dynamics_results/REPORT.md`, `summary.json`.

Time-step sensitivity. The nested noise streams are exact in distribution but couple the paths only approximately while $\widetilde{\Gamma}_h(q)$ moves. The step-refinement differences in Table 4 are comparable to their own standard errors (Fig. 9). They therefore do not resolve the time-discretization error beyond about 10^{-2} to 3×10^{-2}: neither its size nor a convergence order can be inferred from them. What the data do establish is the ordering of the resolved contributions, Krylov ($\leq 10^{-9}$) below spatial ($\leq 6 \times 10^{-7}$) below the combined level of time-step differences and sampling uncertainty (about 10^{-2}). *[Pending: The recomputation repeats these comparisons with the corrected operator and the new ranks.]*

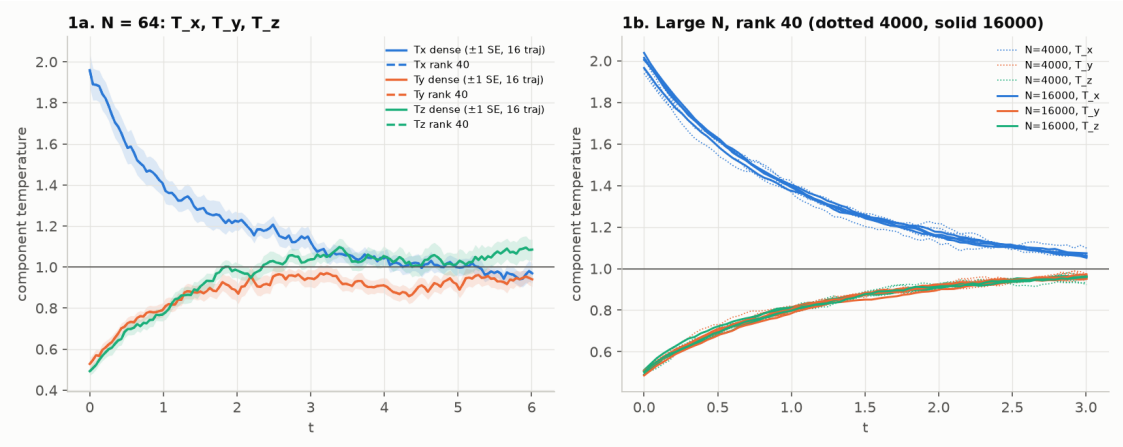


Figure 6: Component temperatures, noise/damping ranks 40/12. (a) $N = 64$, mean over 16 trajectories with ±1 standard error, dense and rank 40. (b) $N = 4000$ and 16000, rank 40. Source: `dynamics_results/plot1_component_temperatures.png`.

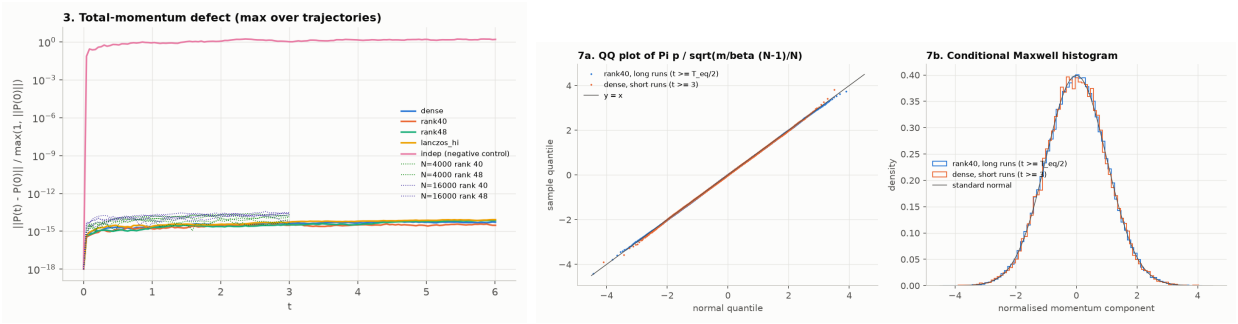


Figure 7: Left: relative total-momentum defect (maximum over trajectories) for all methods, including the control with independent noise. Right: QQ plot and histogram of normalized momenta against the conditional Maxwell law, long runs at $N = 64$. Source: `dynamics_results/plot3, plot7`.

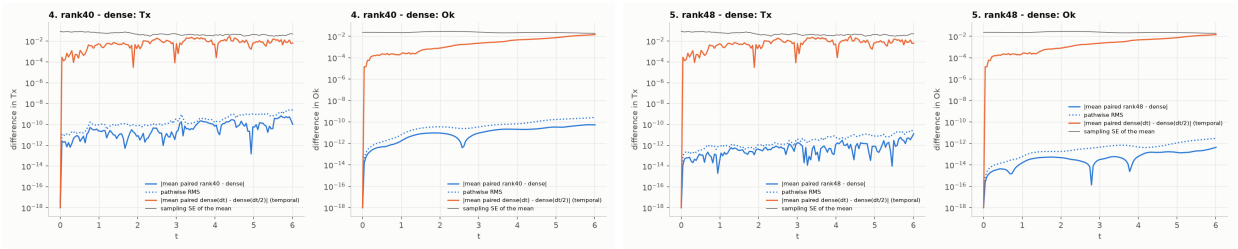


Figure 8: Paired differences of T_x and O_k at $N = 64$: rank 40 minus dense (left) and rank 48 minus dense (right), damping rank 12, with the temporal difference dense($\Delta t/2$)−dense(Δt) and the sampling standard error for comparison. Source: `dynamics_results/plot4, plot5`.

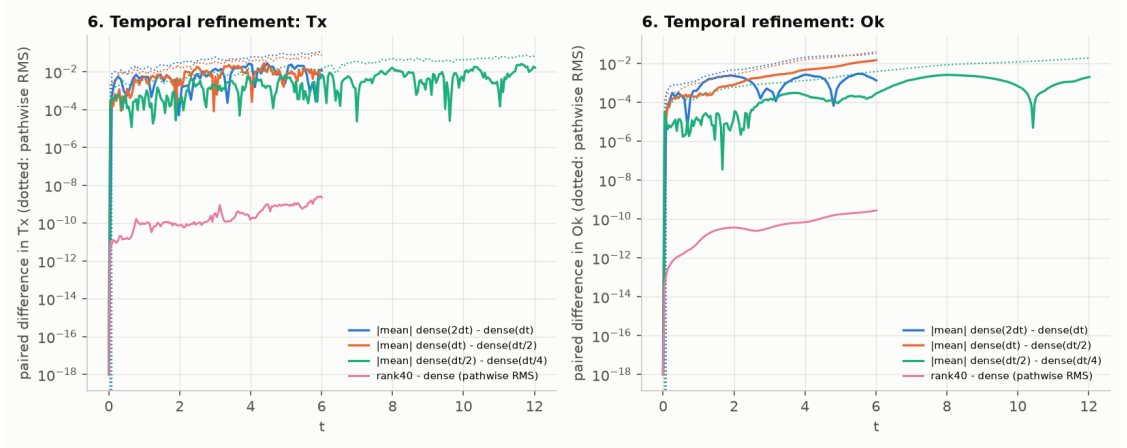


Figure 9: Step refinement at $N = 64$: paired differences between dense runs at $2\Delta t$, Δt, $\Delta t/2$ and $\Delta t/4$, compared with rank 40 minus dense. Source: `dynamics_results/plot6_temporal_refinement.png`.

6.4 Computational cost and scaling

The cost is the measured time of a complete thermostat step at the accuracy targets of Section 6.2: PPPM set $\xi = 0.7$, $s = 4.10$, $\eta_N = 0.7$, $p = 7$, noise rank 40 (noise-action error $\leq 10^{-3}$ through $N = 16000$) and damping rank 16 (damping-action error $\leq 6 \times 10^{-7}$ up to $\Delta t \lambda_{\max} = 20$). The ranks are the same at every size, so each step uses exactly $r_{\text{noise}} + r_{\text{damp}} = 56$ actions of $\widetilde{\Gamma}_h$. Each size was run in a single process with one thread, three warm-up steps and ten timed steps; Table 5 reports medians. An earlier timing with damping rank 12 (52 actions per step, four timed steps, a different session) is superseded and not combined with these numbers.

N	step [s]	setup [s]	Γ_h actions [s]	reorth. [s]	$T/(N \ln N)$ [s]
512	0.789	0.024	0.754	0.003	2.47×10^{-4}
4000	6.35	0.18	6.12	0.027	1.91×10^{-4}
16000	37.4	0.94	36.0	0.12	2.41×10^{-4}

Table 5: Single-process time per complete thermostat step, noise rank 40, damping rank 16 (earlier setting).

About 96% of the step time is spent in the Γ_h actions. The tridiagonal eigensolve and matrix-function evaluation take less than 0.5 ms per step, and projection and other vector operations at most about 0.7%. The normalized time $T(N)/(N \ln N)$ stays between about 1.9×10^{-4} and 2.5×10^{-4} s with no systematic trend (Fig. 10). A descriptive power-law fit to the three points gives $T \propto N^{1.11}$, while $N \ln N$ itself behaves like $N^{1.13}$ over the same range. The measured cost is therefore consistent with $N \log N$ growth for $N = 512$–16000 at these fixed ranks. Since the rank-40 margin is thin at $N = 16000$, a larger N may require a larger noise rank, and the cost per step then grows in proportion to $r_{\text{noise}} + r_{\text{damp}}$.

The timing runs also passed the structural checks: no negative Ritz values (smallest $\approx 1.7 \times 10^{-2}$), total-momentum residual at most 5.9×10^{-15}, and $\|(I - \Pi)\eta\|/\|\eta\| \leq 8.1 \times 10^{-17}$. *[Pending: Equal-accuracy full-step timing with the ranks of the recomputation.]*

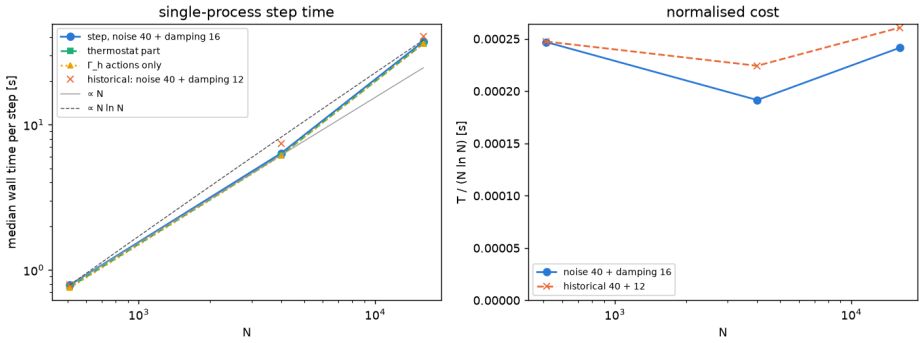


Figure 10: Left: median single-process time per complete step versus N, noise rank 40, damping rank 16, with reference slopes N and $N \ln N$. Right: $T/(N \ln N)$. Source: `dynamics_results/production_timing_rank40_16.png`.

7 Application: Lennard-Jones liquid with kernel A

The experiments of Section 6 use $\kappa = 1$ test configurations and an ideal gas. This section applies the same operator and thermostat to a liquid with a conservative pair potential, at a single state point. All numbers come from the result files of the project repository (`ewald_longitudinal/lj075_results/`, commit `39bb4f6`). Error bars are 95% Student-t intervals over independent replicas unless stated otherwise, and an interval that contains zero is never read as equivalence. Earlier toy studies at a lower density ($N = 64$, $L = 5.5$, $\beta^{-1} = 0.7$) and with a second radial law were also carried out in the project; they are not used here.

7.1 Particle model

Friction kernel. Kernel A is the screened longitudinal kernel of Eq. (1) with an amplitude,

$$K^A(\boldsymbol{r}) = \gamma\, r_{\mathrm{ref}}\, e^{\kappa r_{\mathrm{ref}}}\, K_\kappa(\boldsymbol{r}), \tag{94}$$

chosen so that the radial profile equals γ at $r = r_{\mathrm{ref}}$. We use $\gamma = 0.5$, $\kappa = 0.7$, $r_{\mathrm{ref}} = 1.3$, so the amplitude is $\gamma r_{\mathrm{ref}} e^{\kappa r_{\mathrm{ref}}} = 1.6148$. Between particles the kernel is summed over all periodic images,

$$K_\kappa^{\mathrm{per}}(\boldsymbol{r}) = \sum_{n \in \mathbb{Z}^3} K_\kappa(\boldsymbol{r} + nL), \tag{95}$$

which converges absolutely for $\boldsymbol{r} \notin L\mathbb{Z}^3$ because $\|K_\kappa(\boldsymbol{r})\|_2 = e^{-\kappa r}/r$. Its Fourier coefficients are $V^{-1}\widehat{K}_\kappa(\boldsymbol{k})$, $\boldsymbol{k} \in \mathcal{K}$, and the $k = 0$ coefficient

$$\widehat{K}_\kappa(\boldsymbol{0}) = \int_{\mathbb{R}^3} K_\kappa(\boldsymbol{r})\, d\boldsymbol{r} = \frac{4\pi}{3\kappa^2}\, I \tag{96}$$

is finite because $\kappa > 0$; it is *retained*, i.e. nothing is subtracted from the lattice sum. Eq. (96) follows from $\int_{S^2} P\, d\Omega = \frac{4\pi}{3} I$ and $\int_0^\infty r\, e^{-\kappa r}\, dr = \kappa^{-2}$. The friction matrix is the degree-minus-weight graph of the pair responses

$$(\Gamma(q)X)_i = \sum_{j \neq i} K_{ij}(x_i - x_j), \qquad K_{ij} = \gamma\, r_{\mathrm{ref}}\, e^{\kappa r_{\mathrm{ref}}}\, K_\kappa^{\mathrm{per}}(q_i - q_j). \tag{97}$$

Self images ($j = i$, $n \neq 0$) would act on $x_i - x_i = 0$ and do not enter.

Equations of motion. With positions q, momenta p, mass m and a standard Wiener process W_t in $\mathbb{R}^{3N}$,

$$dq = \frac{p}{m}\, dt, \qquad dp = -\nabla U(q)\, dt - \frac{1}{m}\Gamma(q)p\, dt + \left(2\beta^{-1}\Gamma(q)\right)^{1/2} dW_t, \tag{98}$$

where $U(q) = \sum_{i<j} u(|q_i - q_j|)$ under the minimum-image convention and

$$\begin{aligned} u(r) &= S(r)\, 4\varepsilon \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6} \right], \\ S(r) &= 1 - 10x^3 + 15x^4 - 6x^5, \qquad x = \min\left\{1, \max\left\{0, \frac{r/\sigma - r_{\mathrm{on}}}{r_{\mathrm{cut}} - r_{\mathrm{on}}}\right\}\right\}, \end{aligned} \tag{99}$$

with $r_{\mathrm{on}} = 2$ and $r_{\mathrm{cut}} = 2.5$. The force includes the switch derivative. Reduced units $m = \sigma = \varepsilon = 1$ are used. Since $r_{\mathrm{cut}}\sigma < L/2$, the minimum image is exact for the forces; only the friction uses the full image sum.

Proposition 7.1 (Structure of the friction and invariance)**.** *For pairwise distinct positions:*

1. $\Gamma(q)$ *is symmetric and positive semidefinite, with* $X^{\mathsf{T}}\Gamma X = \sum_{i<j}(x_i - x_j)^{\mathsf{T}} K_{ij}(x_i - x_j)$*;*
2. $\Gamma(q)t = 0$ *for every* $t \in \mathcal{T}$*, so* $\Gamma\Pi = \Pi\Gamma = \Gamma$*;*
3. *the* $k = 0$ *Fourier term contributes* $\gamma r_{\mathrm{ref}} e^{\kappa r_{\mathrm{ref}}} \frac{4\pi}{3\kappa^2 V}(NI_N - \mathbf{1}\mathbf{1}^{\mathsf{T}}) \otimes I_3$ *to* Γ*, which acts on* $\operatorname{Range}\Pi$ *as the multiple* $\gamma r_{\mathrm{ref}} e^{\kappa r_{\mathrm{ref}}} \frac{4\pi N}{3\kappa^2 V}$ *of the identity;*
4. *total momentum is conserved pathwise by Eq. (98);*
5. *the density* $\pi(q,p) \propto \exp\{-\beta[U(q) + |p|^2/(2m)]\}$*, restricted to a fixed total momentum, is stationary for Eq. (98).*

Proof. (1) $K_\kappa(\boldsymbol{r}) = K_\kappa(-\boldsymbol{r}) = K_\kappa(\boldsymbol{r})^{\mathsf{T}}$, so $K_{ij} = K_{ji} = K_{ij}^{\mathsf{T}}$; pairing the (i,j) and (j,i) terms gives the quadratic form. Each image term $e^{-\kappa|\boldsymbol{r}+nL|}|\boldsymbol{r}+nL|^{-1}P(\boldsymbol{r}+nL)$ is positive semidefinite, so $K_{ij} \succeq 0$ and every pair term is nonnegative. (2) Block-row sums of Eq. (97) vanish. (3) Insert the $k = 0$ coefficient $V^{-1}\widehat{K}_\kappa(\mathbf{0})$ of Eq. (96) into Eq. (97); on $\operatorname{Range}\Pi$ the term $\mathbf{1}\mathbf{1}^{\mathsf{T}} \otimes I_3$ vanishes. (4) Pair forces sum to zero, $t^{\mathsf{T}}\Gamma = 0$, and $t^{\mathsf{T}}\Gamma^{1/2} = 0$ because $\Gamma^{1/2}$ has the null space of Γ. (5) The Hamiltonian part preserves π. The friction and noise part of the Fokker–Planck operator is $\nabla_p \cdot [\Gamma(q)(\frac{p}{m}\pi + \beta^{-1}\nabla_p\pi)]$, which vanishes because $\beta^{-1}\nabla_p\pi = -\frac{p}{m}\pi$; there is no spurious drift because Γ depends on q only. □

Positivity in item 1 rests on the pointwise positivity of the real-space kernel, as in Section 2; strict positivity on $\operatorname{Range}\Pi$ is not proved and is checked numerically below. Item 5 concerns the continuous model. Unlike a conservative collision model, $\Gamma p \neq 0$ in general, so kinetic energy is not conserved; the discrete scheme preserves the conditional Maxwell law in its O-step (Appendix F), but its configurational stationary law carries a time-step bias.

The numerical scheme is the splitting $\Psi_{\Delta t}$ of Section 4.4 with the conservative B steps restored and with $\widetilde{\Gamma}_h$ evaluated once per step at the midpoint positions for both damping and noise. The particle–mesh operator of Section 3 is applied to the long-range part including the $k = 0$ mode, and the real-space part sums all pair images within r_c. No stabilization was needed in any test of this section ($\delta_h = 0$ in Eq. (40), so $\Gamma_h = \widetilde{\Gamma}_h$), and the Lanczos evaluation is the direct-start method of Section 4.2 with fixed ranks; a nonpositive Ritz value aborts the run.

7.2 Exact noise coupling across time steps

To compare time steps pathwise, levels Δt and $2\Delta t$ are driven by coupled noise. The finest level draws $\xi \sim \mathcal{N}(0, \Pi)$. A coarse step forms its input from two successive fine inputs ξ_1, ξ_2 with its own friction matrix $\Gamma_c = \widetilde{\Gamma}_h(q^c)$ at its own midpoint positions:

$$\xi_c = a(\Gamma_c)\xi_1 + b(\Gamma_c)\xi_2, \qquad a(\lambda) = \frac{e^{-x}}{\sqrt{1+e^{-2x}}}, \quad b(\lambda) = \frac{1}{\sqrt{1+e^{-2x}}}, \quad x = \frac{\Delta t\,\lambda}{m}. \tag{100}$$

Since $a^2 + b^2 = 1$ and $a(\Gamma_c)$, $b(\Gamma_c)$ commute with Π, $\xi_c \sim \mathcal{N}(0, \Pi)$ conditionally on the past, so every level samples its own correct law. For a frozen matrix, two fine O-steps give $R_{\Delta t}^2 p + R_{\Delta t} f_{\Delta t}\xi_1 + f_{\Delta t}\xi_2$, and $f_{2\Delta t}\xi_c$ equals the noise part because $f_{\Delta t}(\lambda)^2 / f_{2\Delta t}(\lambda)^2 = (1 + e^{-2x})^{-1}$; hence one coarse step equals two fine ones. In the implementation this identity holds to 3×10^{-15} with dense matrix functions and to $\leq 4 \times 10^{-11}$ with Lanczos. Equal seeds alone would not couple the levels.

7.3 State point and runs

The state point is $\rho^* = N\sigma^3/V = 0.75$, $\beta^{-1} = 1$, $N = 256$, $L = 6.98864$. The Ewald–PPPM set is $\xi = 0.85$, $s = 4.1$ ($r_c = 4.824$, $k_c = 6.97$), requested $\eta_N = 0.6779$, B-spline order $p = 8$, $M = 24$ (realised $k_c h/\pi = 0.646$), 1935 retained modes and none excluded. Here $r_c > L/2 = 3.49$, so the real-space sum includes several images per pair. The $k = 0$ shift of Proposition 7.1(3) equals 10.35. An initial set with $p = 7$, cost-optimized at the earlier toy state, gave relative operator errors 2.04–2.15 $\times 10^{-7}$ on lattice-like states at this density, above the budget 2×10^{-7}, so p was raised to 8. Table 6 lists the runs.

experiment	Δt	ranks	initial states / replicas	length, burn-in, output
static verification	–	scan 2–56	42 representative states; 72 + 38 states visited by the runs below (exact O-step inputs)	static
equilibration	0.005	12/5	8 runs: fcc, random sequential addition, canonical, canonical with $\beta^{-1} = 2$ momenta (2 each)	50; burn-in measured 2, recommended 5; every step
time-step study	0.00125, 0.0025, 0.005, 0.01, coupled	12/5 (4)	16 canonical starts	$t_{\text{end}} = 10$; window $[1, 10]$; every step
long chains	0.005, 0.01, coupled	12/5 (4)	11 canonical starts	100; after $t = 10$; every 0.01
operator sensitivity	0.005	12/5	2 replicas, production vs. tighter PPPM set, same noise	10
step cost	0.005	12/5, 14/5	6 interleaved rounds of 40 steps, one thread	–

Table 6: Runs for the Lennard-Jones state point ($N = 256$, $\rho^* = 0.75$, $\beta^{-1} = 1$, PPPM $\xi = 0.85$, $s = 4.1$, $p = 8$, $M = 24$). Canonical starts: fcc or random sequential addition followed by 30–50 time units of scalar Langevin dynamics with conservative forces only. Ranks are noise/damping (coupling).

7.4 Operator accuracy on static and visited states

The reference is the dense full-periodic matrix Γ of Eq. (97), computed by a real-space lattice sum (near images directly, far images by a tensor Chebyshev fit; brute-force self-check 9.4×10^{-16} of the largest entry) with the $k = 0$ term retained. The measured quantity is the relative spectral error $\|\widetilde{\Gamma}_h - \Gamma\|_2/\|\Gamma\|_2$, with budget 2×10^{-7}. Because the exact $k = 0$ shift dominates $\|\Gamma\|_2$ ($\lambda_{\max} \approx 12$), the error is also reported relative to the norm of Γ minus its $k = 0$ term.

Over 152 states—42 representative states, 72 frames visited by the equilibration runs (including frames of minimal pair distance and of extreme Ritz values) and 38 frames of the time-step and long chains—the relative error is at most 7.6×10^{-8}, and at most 1.60×10^{-7} relative to the $k \neq 0$ part (Fig. 11a). Structural checks on all states: symmetry residual $\leq 2 \times 10^{-17}$, null-space residual $\leq 2 \times 10^{-15}$ for both matrices, literal FFT action versus independent dense assembly $\leq 8 \times 10^{-16}$, and smallest eigenvalue on Range Π between 4.58 and 4.84 (Fig. 11c). Two cheaper or tighter PPPM

sets also pass on the 42 representative states ($\eta_N = 0.6$, $p = 7$, $M = 27$: $\leq 7.6 \times 10^{-8}$; $s = 4.4$, $\eta_N = 0.6$, $p = 7$, $M = 30$: $\leq 1.6 \times 10^{-8}$).

Lanczos ranks. The Lanczos error is the relative 2-norm error of the rank-r approximation against the dense matrix function of the same $\widetilde{\Gamma}_h$, for damping and noise functions, over Gaussian and Maxwell vectors, the actual momentum and noise of the step, longitudinal and transverse plane waves at $|\boldsymbol{k}| = 2\pi/L$ and $\sqrt{2}\,2\pi/L$, and slow-mode-dominated vectors. The strict rule requires the overall error and its projections on the $|\boldsymbol{k}| = 2\pi/L$ modes and on the lowest 3 and 12 eigenmodes to be at most one tenth of the operator budget. Rank 12 suffices on the representative states, but on one visited state (a melting fcc start at $t = 5$) it leaves 2.88×10^{-8} in the $|\boldsymbol{k}| = 2\pi/L$ projection, driven by a transverse plane wave. The recommendation is therefore $r_{\text{noise}} = 14$, $r_{\text{damp}} = 5$, with worst errors 6.4×10^{-10} and 2.8×10^{-11} at $\Delta t = 0.005$ over all 152 states (Fig. 11b). These ranks are far below the 40/16 of the earlier setting of Section 6.2, which were set for the ideal gas with a 10^{-3} root-action budget up to $N = 16000$; the present spectrum is narrow ($\lambda \in [4.6, 12.5]$ on Range Π, largely because of the $k = 0$ shift). The dynamic runs below used 12/5; their on-line monitor on the actual inputs never exceeded 1.6×10^{-10} overall and 1.8×10^{-9} in the $|\boldsymbol{k}| = 2\pi/L$ projection for the noise.

Physical effect of the operator error. Two replicas were run in lock-step with the same noise and start, once with the production set and once with the tighter set (relative error $\leq 1.6 \times 10^{-8}$), at $\Delta t = 0.005$ for 10 time units. The largest differences of time averages were 5.9×10^{-5} (kinetic temperature, relative), 1.4×10^{-5} (U/N), 1.3×10^{-4} (mean-square displacement at $t = 0.5$–5) and 4.5×10^{-4} (pressure, absolute). With two replicas these are observed maxima, not confidence intervals.

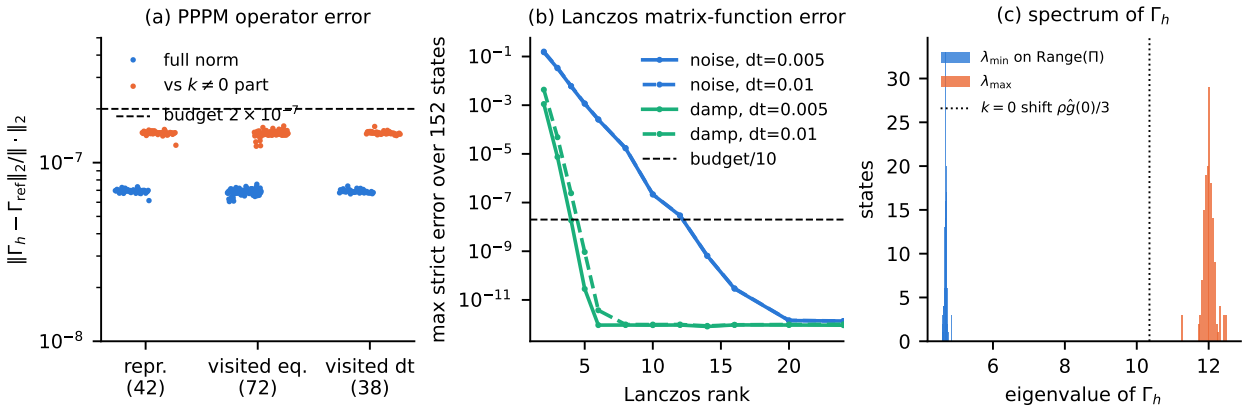


Figure 11: Static verification at the Lennard-Jones state point (kernel A, $N = 256$, $\rho^* = 0.75$, $\beta^{-1} = 1$, PPPM $\xi = 0.85$, $s = 4.1$, $p = 8$, $M = 24$), 152 configurations; each point is one configuration, no error bars. (a) Relative spectral error of $\widetilde{\Gamma}_h$ against the full-periodic matrix, full norm and relative to the $k \neq 0$ part; dashed: budget. (b) Strict Lanczos error (maximum over probe types and states) versus rank; dashed: budget/10. (c) Extreme eigenvalues on Range Π; dotted: $k = 0$ shift 10.35. Data: `verify_{representative,visited_eq,visited_dt}_A.json`.

7.5 Time-step dependence

Four levels $\Delta t \in \{0.01, 0.005, 0.0025, 0.00125\}$ were integrated in lock-step with the coupling of Eq. (100) from 16 independent canonical starts to $t_{\text{end}} = 10$. Differences are paired by replica; time averages use $[1, 10]$. The acceptance criteria were committed before any run: 1% relative for kinetic temperature, U/N, the first peak of the radial distribution function and the mean-square displacement at $t = 0.1, 0.5, 1, 2, 5$; 0.0075 absolute for pressure; 1% for a weighted norm of the radial-distribution difference; 1% of the initial value for the maximum velocity-autocorrelation difference up to $t = 2$. PASS requires the whole 95% interval inside the band, FAIL the whole interval outside, otherwise INCONCLUSIVE. The decision rule was to accept the largest Δt whose criteria all PASS.

criterion	0.0025	0.005	0.01
kinetic temperature	-0.11% $[-0.32, 0.09]$ P	-0.12% $[-0.30, 0.07]$ P	-0.52% $[-0.78, -0.26]$ P
U/N	-0.12% $[-0.23, -0.01]$ P	-0.05% $[-0.14, 0.04]$ P	-0.16% $[-0.29, -0.03]$ P
pressure (abs.)	-0.026 $[-0.049, -0.003]$ I	0.0006 $[-0.018, 0.019]$ I	-0.024 $[-0.050, 0.002]$ I
RDF norm, registered bound	1.78% I	1.68% I	1.82% I
RDF first-peak height	0.01% $[-0.44, 0.47]$ P	-0.05% $[-0.50, 0.41]$ P	0.02% $[-0.60, 0.64]$ P
RDF first-peak position	0.03% $[-0.22, 0.28]$ P	0.00% $[-0.23, 0.22]$ P	-0.04% $[-0.23, 0.16]$ P
VACF, bounds	$[0, 0.39\%]$ P	$[0.004\%, 0.41\%]$ P	$[0.26\%, 0.78\%]$ P
MSD(0.1)	-0.03% $[-0.28, 0.22]$ P	0.04% $[-0.19, 0.27]$ P	0.14% $[-0.20, 0.48]$ P
MSD(0.5)	0.13% $[-0.23, 0.50]$ P	0.41% $[-0.36, 1.18]$ I	0.47% $[-0.45, 1.40]$ I
MSD(1)	0.25% $[-0.40, 0.90]$ P	0.52% $[-0.63, 1.67]$ I	0.87% $[-0.54, 2.28]$ I
MSD(2)	0.69% $[-0.59, 1.98]$ I	0.89% $[-1.08, 2.85]$ I	1.80% $[-0.15, 3.75]$ I
MSD(5)	3.44% $[1.03, 5.86]$ **F**	3.08% $[-0.20, 6.36]$ I	2.99% $[-0.32, 6.31]$ I

Table 7: Pre-registered time-step comparison against $\Delta t = 0.00125$ (16 coupled replicas, ranks 12/5; mean [95% CI], relative unless stated). P/I/F = PASS/INCONCLUSIVE/FAIL. Data: `dt_study.json`.

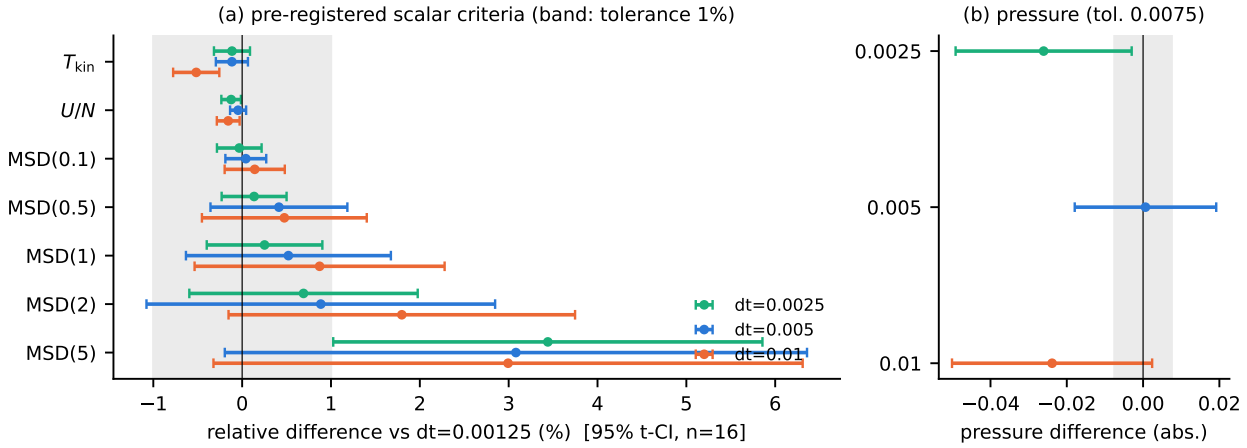


Figure 12: Scalar time-step criteria of Table 7: each level against $\Delta t = 0.00125$, 16 coupled replicas, paired 95% Student-t intervals; grey bands are the registered tolerances.

Under the registered rule *no time step is validated* with 16 replicas (Table 7, Fig. 12). The finest-two comparison is not all-PASS, and no coarser level passes every criterion; pressure, the RDF norm and the mean-square displacement for $t \geq 0.5$ are undecided at every level. On the

criteria that are decidable at this sample size, all three coarse levels PASS; a rule restricted to that subset would select $\Delta t = 0.01$, but it is post hoc and is not used as validation. At $\Delta t = 0.01$ the kinetic-temperature (-0.52%) and U/N (-0.16%) biases are resolved, though within 1%, and the velocity-autocorrelation difference bound lies above its noise floor; at $\Delta t = 0.005$ neither bias is resolved. We therefore take $\Delta t = 0.005$ as a *provisional* step and regard $\Delta t = 0.01$ as acceptable for static averages with the quoted biases.

The registered MSD(5) verdict at $\Delta t = 0.0025$ is FAIL. A post-hoc comparison shows the same $\approx 3\%$ offset for every coarse level while the coarse levels agree with one another (0.005 vs. 0.0025: -0.35% $[-3.6, 2.9]$), which points to a fluctuation of the reference level; this reading is post hoc and does not change the verdict. Pairing loses its benefit after $t \approx 1$–2: the median RMS velocity difference to the finest path is 0.025, 0.057, 0.12 at $t = 1$ for the three coarse levels and 0.80–0.86 at $t = 5$ (Fig. 17). A 1% verdict would need about 25 replicas for MSD(0.5), 80 for MSD(1) and 60–100 for the pressure tolerance.

Diffusion. Eleven coupled two-level chains ($\Delta t = 0.005$ and 0.01, 100 time units, analysed after $t = 10$) give windowed mean-square-displacement diffusion coefficients (Fig. 13). With a plateau rule fixed before these runs were analysed (paired change between consecutive windows within $\pm 2\%$ over the whole interval), no plateau is confirmed: the coefficient decreases by 1.9% $[1.2, 2.7]$ and then 1.1% $[0.2, 2.0]$ per window doubling up to $[4, 8]$, and later steps have intervals of ± 1.4–2.6%. The best estimate is $D \approx 0.0284$–0.0286 at $\Delta t = 0.005$, not a converged long-time value. The time-step effect is resolved only in the shortest window ($+0.80\%$ $[0.05, 1.55]$).

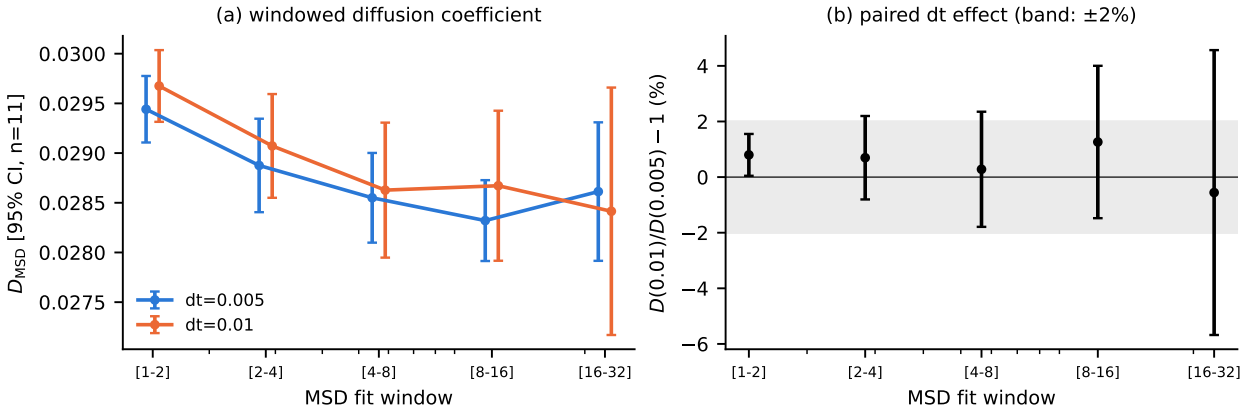


Figure 13: (a) Windowed mean-square-displacement diffusion coefficient, 11 coupled chains of 100 time units, ranks 12/5, 95% intervals over chains. (b) Paired relative difference between $\Delta t = 0.01$ and 0.005; grey band: $\pm 2\%$ plateau tolerance. Data: `longD.json`.

7.6 Equilibration and static quantities

Eight runs from four start types give, for $t \geq 25$, $U/N = -4.6697$, kinetic temperature 1.0003, pressure 0.770 and first RDF peak 2.558 (Fig. 18). The kinetic temperature is $|p - \bar{p}|^2/(3m(N-1))$ and the pressure uses the conservative virial only. A block rule fixed before the analysis gives a burn-in of 2 time units; 5 are recommended. Start-type dependence after burn-in was not detected (one-way ANOVA $p = 0.21$–0.78), but with two runs per type initial-state independence is under-sampled. Velocity autocorrelation and Green–Kubo integrals require output at every step at

$\Delta t = 0.005$ (quadrature bias 0.23% at interval 0.01 against a 0.2% rule), whereas the mean-square displacement is unaffected up to interval 0.1.

7.7 Dynamical correlation functions

With $v_i = p_i/m$ and time origins t_0, the observables are

$$Z(t) = \frac{\langle v_i(t_0) \cdot v_i(t_0+t)\rangle}{\langle |v|^2\rangle}, \qquad C_b(t) = \frac{\langle v_i(t_0) \cdot v_j(t_0+t)\rangle_{|q_i(t_0)-q_j(t_0)|\in b}}{\langle |v|^2\rangle}, \tag{101}$$

$$j(\boldsymbol{k},t) = \frac{1}{N}\sum_i v_i(t) e^{i\boldsymbol{k}\cdot q_i(t)}, \qquad C_{L,T}(k,t) = \frac{\langle j_{L,T}(\boldsymbol{k},t_0+t)\,\overline{j_{L,T}(\boldsymbol{k},t_0)}\rangle}{\langle |j_{L,T}(\boldsymbol{k},t_0)|^2\rangle}, \tag{102}$$

$$g_d(r,t) = \frac{V}{N(N-1)}\left\langle \sum_{i\neq j} \frac{\delta\big(r - |q_j(t_0+t) - q_i(t_0)|\big)}{4\pi r^2}\right\rangle, \tag{103}$$

where j_L and j_T are the components of j along and across $\boldsymbol{k}$. The shells b are $[0, 1.59)$, $[1.59, 2.55)$ and $[2.55, L/2)$, the first and second minima of the radial distribution function; C_L, C_T are averaged over the modes with $|\boldsymbol{k}| = 0.899,\ 1.271,\ 1.798$; g_d is evaluated at $t = 0.1, 0.5, 1, 2$. Zero total momentum gives $\sum_{j\neq i} v_i \cdot v_j = -|v_i|^2$, so each C_b starts near $-1/(N-1)$; nothing is subtracted.

Two groups are analysed separately and never pooled: the 16 four-level chains (origins $t_0 \in [1, 8]$, lags ≤ 2) and the 11 long chains (only $\Delta t = 0.01$ vs. 0.005, origins $t_0 \in [10, 98]$). Curves are normalized by each level's own $t = 0$ value. For a window of a curve with paired mean difference $\overline{\Delta}(t)$, the reference mean curve is cut into sign lobes and

$$E_{\text{sig}} = \sup_{t\in\text{window}} \frac{|\overline{\Delta}(t)|}{S(t)}, \tag{104}$$

where $S(t)$ is the amplitude of the lobe containing t. Difference bands are simultaneous at 97.5% over the window (maximum of a sign-flip max-t and Bonferroni-t constant), lobe amplitudes simultaneous at 97.5%, jointly $\geq 95\%$. A window PASSes if the upper bound of E_{sig} is $\leq 1\%$ and every lobe in it is resolved to $\pm 10\%$, FAILs if the lower bound exceeds 1%, and is INCONCLUSIVE otherwise. The protocol was frozen and amended once (each lobe judged on its own) before any difference was computed.

Results at 1% of the lobe amplitude (Figs. 14, 15; second group in Appendix H):

- Of 71 observable-window cells, one supports display at $\Delta t = 0.01$ (and 0.005): a single g_d bin at $r = 0.84$, $t = 0.1$, where $g_d \approx 0.012$; it says nothing about the van Hove structure. Five cells are not supported because of FAILs and 65 are INCONCLUSIVE.

- Resolved deviations at $\Delta t = 0.01$ vs. 0.00125: $\langle |v|^2\rangle$ lower by 0.38% $[0.09, 0.67]$ (PASS); slower decay of C_T at $|\boldsymbol{k}| = 0.899$ (half-decay time +3.6% $[0.6, 6.6]$); $g_d(r < 0.9, t = 2)$ with $E_{\text{sig}} = 1.9\%$ $[0.14, 4.6]$; all but the first are INCONCLUSIVE against 1%.

- Seven of 284 cells FAIL, six involving $\Delta t = 0.01$ and five in the 0.01 vs. 0.005 group; all lie in windows whose reference lobes are unresolved, and at this sample size the counts cannot separate chance from a time-step effect.

- No whole-curve deviation is resolved in the four-level group, which is not equivalence: the simultaneous upper bounds at $\Delta t = 0.01$ are 1.0×10^{-2} of $Z(0)$, 3.2×10^{-3} of $\langle |v|^2\rangle$ for the first VCCF shell, 0.092 and 0.068 for C_L, C_T at $|\boldsymbol{k}| = 0.899$, and 0.030, 0.045 for g_d at $t = 1, 2$.

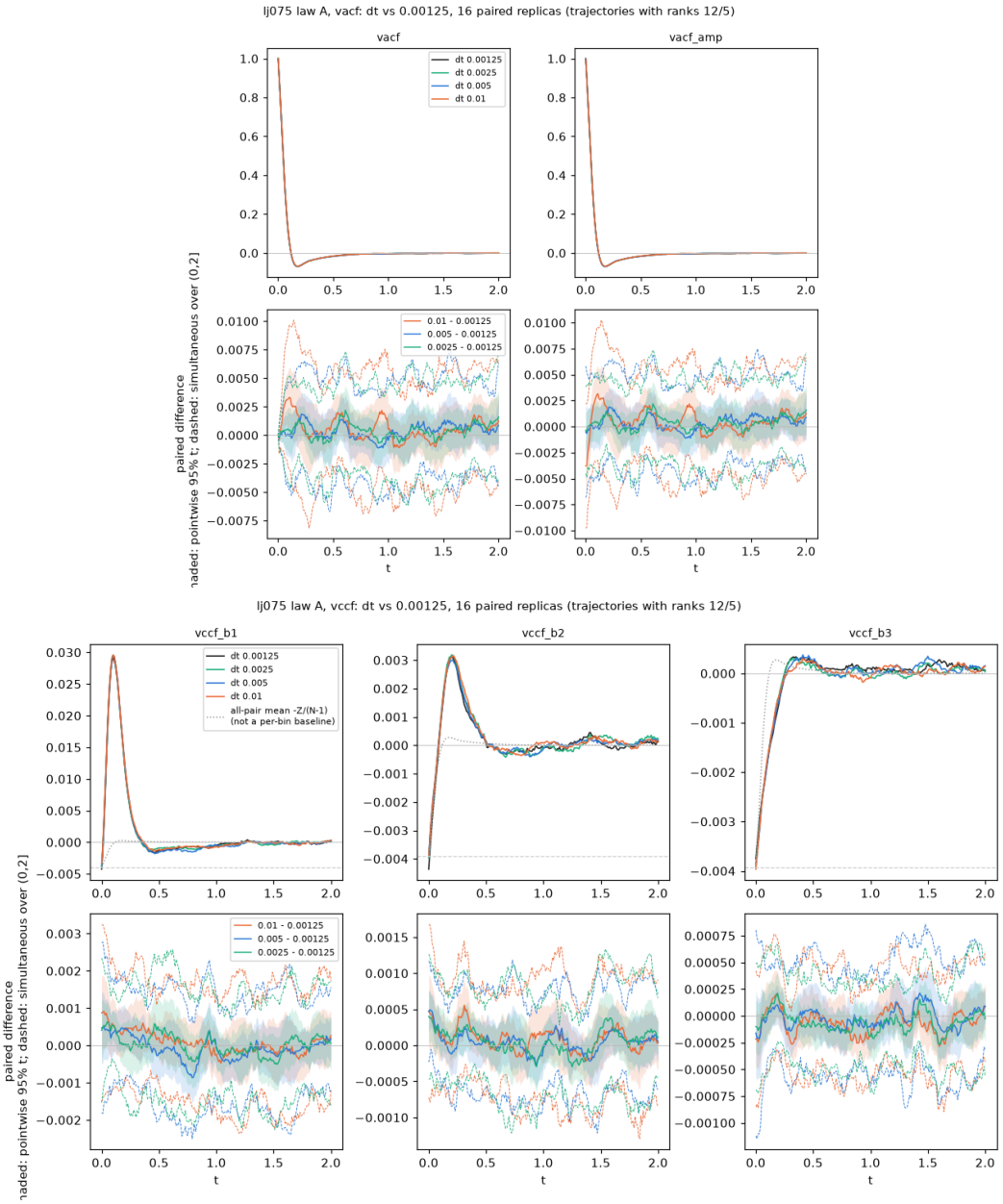


Figure 14: $Z(t)$ (top) and $C_b(t)$ for the three shells (bottom), 16 coupled four-level replicas, ranks 12/5. Upper rows: level means; lower rows: paired differences against $\Delta t = 0.00125$; shading: pointwise 95% interval (not used for verdicts); dashed: simultaneous 97.5% band over $(0, 2]$. Dotted line: all-pair mean $-Z/(N-1)$.

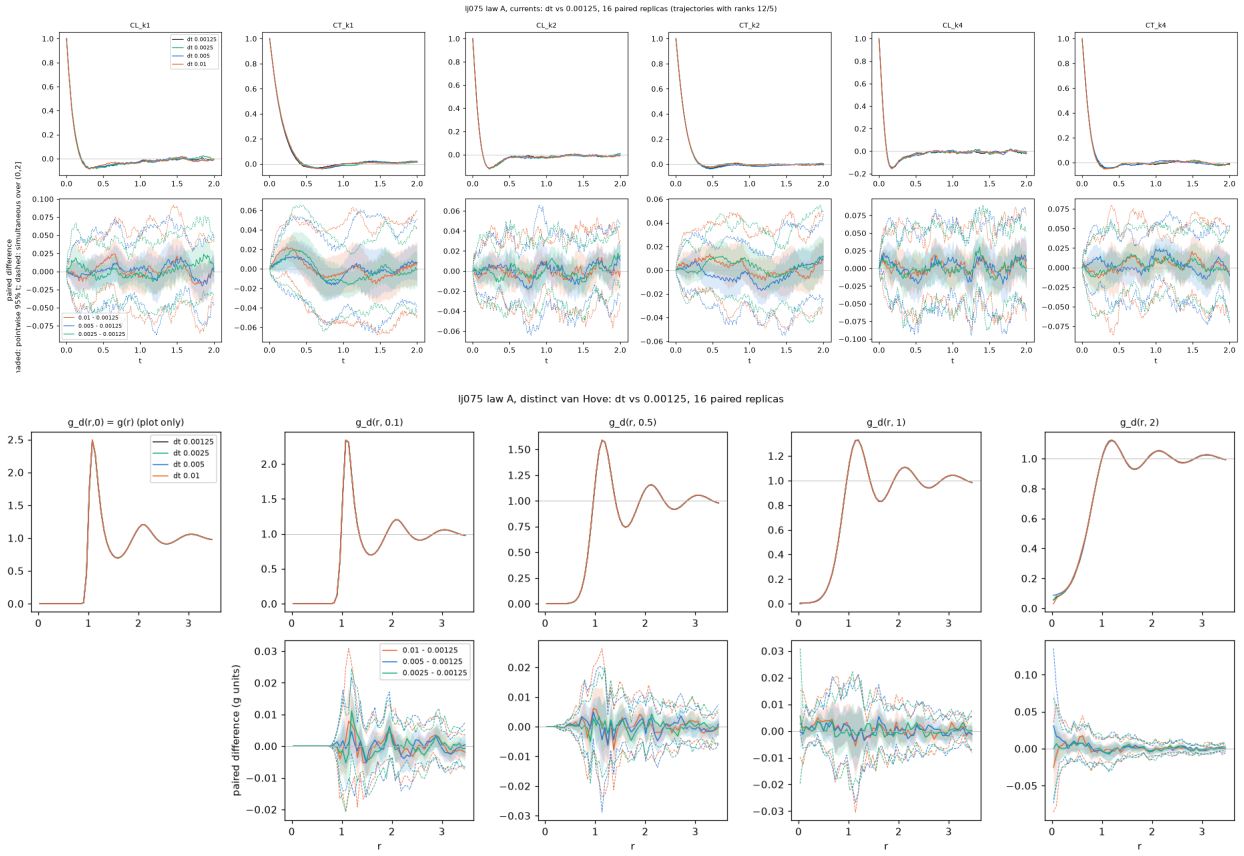


Figure 15: C_L, C_T at $|\boldsymbol{k}| = 0.899, 1.271, 1.798$ (top) and $g_d(r,t)$ at $t = 0, 0.1, 0.5, 1, 2$ (bottom; $t = 0$ shown only); data and bands as in Fig. 14. The g_d band covers only bins with at least 100 reference pairs per replica.

- Most cells are undecided because weak reference lobes are not resolved to ±10%, or because the difference bands reach 1.5–8.5% of the lobe amplitude.

Decay of a correlation function to zero is not used as evidence of model correctness, and long-lag differences are not attributed to a mechanism.

7.8 Cost

All timings in this subsection come from one virtual machine (one thread) and are not combined with the timings of Section 6. The median CPU time of a full step at $\Delta t = 0.005$ (forces, operator setup, damping and noise) was 124.8 ms at ranks 12/5 and 134.9 ms at 14/5, i.e. 0.0075 core-hours per time unit at 14/5; the two alternative PPPM sets took 117.9 and 140.0 ms. In a separate session, $\Delta t = 0.01$ saved 48.7% (12/5) and 50.4% (14/5) per physical time relative to $\Delta t = 0.005$. A local single-path run of the cluster scripts (one state, $t \leq 1$, shared noise, not run on a cluster) gave median steps of 0.208 s with PPPM and Lanczos 14/5 versus 1.129 s with the dense full-periodic matrix and eigendecomposition, peak memory 121 versus 604 MB; the velocity difference at $t = 1$ was 7.4×10^{-8} of the RMS velocity, dominated by the operator error (Appendix H). No size scan was performed at this state point.

7.9 Scope

At this state point the following are established: the operator budget and positivity on Range Π on 152 static and visited states; the Lanczos ranks 14/5 under the strict rule; momentum conservation to round-off; a negligible effect of the operator error on time averages in two replicas; and 1% agreement of temperature, U/N, the first RDF peak, the velocity autocorrelation for $t \leq 2$ and MSD(0.1) at all three coarse time steps. Provisional or under-sampled: the time step itself, the time-step effect on the RDF norm, MSD($t \geq 0.5$), pressure and diffusion, the diffusion plateau, initial-state independence, and most 1% comparisons of the correlation functions. Not addressed: other N, a convergence theorem for the scheme with configuration-dependent friction and Lanczos noise, an optimized influence function, a cluster run, and comparison with molecular dynamics or experiment. Raw trajectories of these runs are not in the repository; all figures and tables above are regenerated from cached analysis files.

8 Conclusion

Sections 2–5 address the consequences of the screened longitudinal Ewald long-range kernel for the particle graph and its PPPM discretization. The analysis rests on two properties: the real-space long-range pair kernel is pointwise positive semidefinite, and its Fourier multiplier has Gaussian decay $\mathcal{O}(e^{-k^2/(4\xi^2)})$. The Fourier multiplier itself need not be positive semidefinite; positivity of the exact friction follows instead from the pointwise semidefinite pair kernel and the graph degree-minus-weight structure.

Finite reciprocal and particle–mesh approximations preserve symmetry and the translation nullspace when adjoint spreading/gathering and a common degree-minus-weight construction are used. Their operator error is controlled by the complete pair-response discrepancy. For ideal particle–mesh deconvolution, the leading mesh error is generated by the linear off-grid alias cross terms and satisfies

$$\epsilon_{\text{mesh}} = \mathcal{O}(h^p) \tag{105}$$

for a B-spline assignment of order p. The resulting practical error model is

$$\epsilon_{\mathrm{op}} \lesssim C_E e^{-s^2} + C_M(\xi h)^p, \qquad s = \xi r_c = \frac{k_c}{2\xi}. \tag{106}$$

A useful initial choice is

$$s = \sqrt{\log(2/\mathrm{tol})}, \qquad r_c = \frac{s}{\xi}, \qquad k_c = 2\xi s, \tag{107}$$

followed by a mesh chosen from the measured envelope of C_M and a Nyquist safety condition. Accuracy is certified by a complete pair-response operator bound, while

$$\epsilon_{\mathrm{op}} < \lambda_* \tag{108}$$

provides a sufficient, although conservative, certificate that the raw discrete graph remains positive semidefinite.

The experiments of Section 6, obtained in the earlier setting described there, support the following conclusions.

1. On the tested configurations the mesh error is carried by the linear alias cross terms, the fitted mesh slopes are consistent with the h^p law, and no negative graph eigenvalue occurred (Section 6.1).

2. At a fixed high accuracy 10^{-8} the direct-start Lanczos rank grows with N, approximately as $N^{0.34}$, so a fixed-rank statement at that accuracy is not supported; the range start is more expensive.

3. Under the production budgets (10^{-3} for the noise action, 10^{-6} for the damping action) the fixed ranks 40 and 16 sufficed through $N = 16000$, with a thin noise margin at the largest size (Section 6.2).

4. The ideal-gas dynamics preserve total momentum pathwise to round-off, relax anisotropic temperatures toward equipartition, and reproduce the conditional Maxwell law in the long $N = 64$ runs. Krylov differences on the same $\widetilde{\Gamma}_h$ ($\leq 10^{-9}$) are below the spatial proxy ($\leq 6 \times 10^{-7}$), which is below the combined level of step-refinement differences and sampling uncertainty (about 10^{-2}); the time discretization error itself is not resolved (Section 6.3).

5. With ranks 40/16 a complete step uses 56 actions of $\widetilde{\Gamma}_h$, and its measured cost is consistent with $N \log N$ growth for $N = 512$–16000 (Section 6.4).

The large-N runs verify relaxation and coupled fixed-rank accuracy but do not establish large-N equilibrium. The dynamic runs used damping rank 12, the timings damping rank 16. Production ranks for the corrected operator are determined by the recomputation in progress, and the ranks above should not be transferred to it. The experiments of Section 6 contain no conservative potential; dynamics with a Lennard-Jones potential are treated separately in Section 7.

Section 7 extends the validation to a Lennard-Jones liquid with the amplitude-scaled kernel A at $\rho^* = 0.75$, $\beta^{-1} = 1$, $N = 256$. There the particle–mesh operator meets a 2×10^{-7} relative spectral budget against the full-periodic matrix with the $k = 0$ term retained on 152 static and visited configurations, remains positive on $\mathrm{Range}\,\Pi$ without stabilization, and Lanczos ranks $r_{\mathrm{noise}} = 14$, $r_{\mathrm{damp}} = 5$ suffice under a strict error rule; these ranks are specific to that state point and accuracy budget. A pre-registered, pathwise-coupled time-step study validates no time step at 16 replicas: $\Delta t = 0.005$ is provisional, and $\Delta t = 0.01$ shows resolved sub-percent biases in temperature and potential energy. The long-time diffusion plateau and most 1% comparisons of the dynamical correlation functions remain under-sampled.

A Gaussian-scale representation

This appendix proves Proposition 2.1.

Gaussian representations of the two factors. We factor the scalar part of the kernel as

$$\frac{e^{-\kappa r}}{r^3} = \frac{e^{-\kappa r}}{r} \cdot \frac{1}{r^2} \tag{109}$$

and represent each factor as a superposition of Gaussians in r. For the Yukawa factor we use the standard representation

$$\frac{e^{-\kappa r}}{r} = \frac{1}{\sqrt{\pi}} \int_0^\infty t^{-1/2} e^{-\kappa^2/(4t)} e^{-tr^2}\, dt, \tag{110}$$

which is the integral $\int_0^\infty t^{-1/2} e^{-at-b/t}\, dt = \sqrt{\pi/a}\, e^{-2\sqrt{ab}}$, $a, b > 0$, evaluated at $a = r^2$ and $b = \kappa^2/4$. For the second factor,

$$\frac{1}{r^2} = \int_0^\infty e^{-ur^2}\, du. \tag{111}$$

Thus the Gaussian-scale weights of the two factors are $\pi^{-1/2} t^{-1/2} e^{-\kappa^2/(4t)}$ and the constant 1.

Convolution of the weights. Multiplying the two representations, the Gaussians combine as $e^{-tr^2} e^{-ur^2} = e^{-(t+u)r^2}$, so the product is again a superposition of Gaussians, with scale $s = t + u$. Its weight is the Laplace convolution of the two weights. Explicitly, by Tonelli's theorem, since all integrands are nonnegative,

$$\begin{aligned}
\frac{e^{-\kappa r}}{r^3} &= \frac{1}{\sqrt{\pi}} \int_0^\infty \int_0^\infty t^{-1/2} e^{-\kappa^2/(4t)} e^{-(t+u)r^2}\, du\, dt && (112)\\
&= \frac{1}{\sqrt{\pi}} \int_0^\infty \int_t^\infty t^{-1/2} e^{-\kappa^2/(4t)} e^{-sr^2}\, ds\, dt && (113)\\
&= \int_0^\infty \left[\frac{1}{\sqrt{\pi}} \int_0^s t^{-1/2} e^{-\kappa^2/(4t)} \cdot 1\, dt \right] e^{-sr^2}\, ds && (114)\\
&= \int_0^\infty \rho_\kappa(s) e^{-sr^2}\, ds, && (115)
\end{aligned}$$

where the second line substitutes $s = t + u$ in the inner integral and the third exchanges the order of integration over $0 < t < s < \infty$. The bracket is the convolution $\int_0^s (\pi^{-1/2} t^{-1/2} e^{-\kappa^2/(4t)}) \cdot 1\, dt$ of the Yukawa weight with the constant weight of $1/r^2$. Multiplying by $\boldsymbol{r}\boldsymbol{r}^{\mathsf{T}}$ and using

$$K_\kappa(\boldsymbol{r}) = e^{-\kappa r} \frac{\boldsymbol{r}\boldsymbol{r}^{\mathsf{T}}}{r^3} \tag{116}$$

gives Eq. (2).

Closed form of the weight. Let $\varphi(s)$ denote the right-hand side of the second equality in Eq. (3). Differentiation gives

$$\begin{aligned}
\frac{d}{ds}\left[\frac{2\sqrt{s}}{\sqrt{\pi}} e^{-\kappa^2/(4s)} \right] &= \frac{1}{\sqrt{\pi}} \left(s^{-1/2} + \frac{\kappa^2}{2} s^{-3/2} \right) e^{-\kappa^2/(4s)}, && (117)\\
\frac{d}{ds}\left[\kappa \left[\operatorname{erf}\left(\frac{\kappa}{2\sqrt{s}} \right) - 1 \right] \right] &= -\frac{\kappa^2}{2\sqrt{\pi}}\, s^{-3/2} e^{-\kappa^2/(4s)}, && (118)
\end{aligned}$$

so $\varphi'(s) = \pi^{-1/2} s^{-1/2} e^{-\kappa^2/(4s)}$. Since $\varphi(s) \to 0$ as $s \to 0^+$, φ coincides with the integral in Eq. (3).

Properties of the weight. The integrand in Eq. (3) is positive and bounded by $\pi^{-1/2}t^{-1/2}$. Hence ρ_κ is positive and nondecreasing on $(0,\infty)$, and

$$\rho_\kappa(s) \le \frac{2\sqrt{s}}{\sqrt{\pi}}. \tag{119}$$

In particular, $\rho_\kappa(\xi^2) > 0$.

B Closed-form real-space coefficients

Let

$$z = \frac{\kappa}{2\xi}, \tag{120}$$

and define

$$c_{\kappa,\xi} = \rho_\kappa(\xi^2) = \frac{2\xi}{\sqrt{\pi}} e^{-z^2} + \kappa\left[\operatorname{erf}(z) - 1\right]. \tag{121}$$

The long-range scalar coefficient can be written as

$$\theta_L(r) = Y_\kappa^L(r) - c_{\kappa,\xi} e^{-\xi^2 r^2}, \tag{122}$$

where

$$Y_\kappa^L(r) = \frac{1}{2r}\left[e^{-\kappa r}\operatorname{erfc}(z-\xi r) - e^{\kappa r}\operatorname{erfc}(z+\xi r)\right]. \tag{123}$$

Similarly,

$$\theta_S(r) = Y_\kappa^S(r) + c_{\kappa,\xi} e^{-\xi^2 r^2}, \tag{124}$$

with

$$Y_\kappa^S(r) = \frac{1}{2r}\left[e^{-\kappa r}\operatorname{erfc}(\xi r - z) + e^{\kappa r}\operatorname{erfc}(\xi r + z)\right]. \tag{125}$$

Since

$$Y_\kappa^S(r) + Y_\kappa^L(r) = \frac{e^{-\kappa r}}{r}, \tag{126}$$

the identity $K_\kappa^S + K_\kappa^L = K_\kappa$ follows immediately.

C Localization estimates

This appendix proves Proposition 2.2 and Eq. (17).

Real-space decay. Fix $r > 0$. Since $r^2 e^{-sr^2} = -\frac{d}{ds} e^{-sr^2}$, integration by parts in

$$\theta_S(r) = r^2 \int_{\xi^2}^{\infty} \rho_\kappa(s) e^{-sr^2}\, ds \tag{127}$$

gives

$$\theta_S(r) = \rho_\kappa(\xi^2) e^{-\xi^2 r^2} + \int_{\xi^2}^{\infty} \rho_\kappa'(s) e^{-sr^2}\, ds. \tag{128}$$

The boundary term at infinity vanishes: by Eq. (119), $0 \le \rho_\kappa(s) e^{-sr^2} \le 2\pi^{-1/2}\sqrt{s}\, e^{-sr^2} \to 0$ as $s \to \infty$. By Appendix A,

$$\rho_\kappa'(s) = \frac{1}{\sqrt{\pi s}}\, e^{-\kappa^2/(4s)}, \tag{129}$$

so $0 \le \rho_\kappa'(s) \le (\sqrt{\pi}\,\xi)^{-1}$ for $s \ge \xi^2$. Inserting this bound in the last integral of Eq. (128) and using $\int_{\xi^2}^{\infty} e^{-sr^2}\,ds = e^{-\xi^2 r^2}/r^2$ gives

$$0 \le \theta_S(r) - \rho_\kappa(\xi^2)e^{-\xi^2 r^2} \le \frac{e^{-\xi^2 r^2}}{\sqrt{\pi}\,\xi r^2}. \tag{130}$$

Hence, for fixed $\kappa, \xi > 0$,

$$\theta_S(r) = e^{-\xi^2 r^2}\left[\rho_\kappa(\xi^2) + \mathcal{O}(r^{-2})\right], \qquad r \to \infty. \tag{131}$$

Since $\rho_\kappa(\xi^2) > 0$, this gives $\theta_S(r) \sim \rho_\kappa(\xi^2)e^{-\xi^2 r^2}$, and $K_\kappa^S(\boldsymbol{r}) = \theta_S(r)P(\boldsymbol{r})$ yields Eq. (10).

Fourier transform. For a Gaussian,

$$\mathcal{F}[e^{-sr^2}] = \left(\frac{\pi}{s}\right)^{3/2} e^{-k^2/(4s)}. \tag{132}$$

With the convention used here, multiplication by r_a corresponds to $i\,\partial/\partial k_a$ in Fourier space, so $\mathcal{F}[r_a r_b e^{-sr^2}] = -\partial_{k_a}\partial_{k_b}\mathcal{F}[e^{-sr^2}]$. Since $\partial_{k_a}\partial_{k_b}e^{-k^2/(4s)} = \big(k_a k_b/(4s^2) - \delta_{ab}/(2s)\big)e^{-k^2/(4s)}$,

$$\mathcal{F}[r_a r_b e^{-sr^2}] = \left(\frac{\pi}{s}\right)^{3/2} e^{-k^2/(4s)}\left[\frac{\delta_{ab}}{2s} - \frac{k_a k_b}{4s^2}\right]. \tag{133}$$

Integrating against $\rho_\kappa(s)$ over $(0, \xi^2)$, which is permitted because $\rho_\kappa(s)s^{-7/2}$ is integrable near $s = 0$, gives Eq. (11) with the stated $A_{\kappa,\xi}$ and $B_{\kappa,\xi}$. Differentiating $A_{\kappa,\xi}$ under the integral sign, $\partial_k e^{-k^2/(4s)} = -(k/(2s))e^{-k^2/(4s)}$, gives

$$B_{\kappa,\xi}(k) = \frac{1}{k}\frac{d}{dk}A_{\kappa,\xi}(k), \qquad k > 0. \tag{134}$$

Endpoint asymptotics. Let $k > 0$ and, for $m \in \{5/2, 7/2\}$,

$$I_m(k) = \int_0^{\xi^2} \rho_\kappa(s)s^{-m}e^{-k^2/(4s)}\,ds, \tag{135}$$

so that $A_{\kappa,\xi}(k) = \frac{\pi^{3/2}}{2}I_{5/2}(k)$ and $B_{\kappa,\xi}(k) = -\frac{\pi^{3/2}}{4}I_{7/2}(k)$. Since

$$\frac{d}{ds}e^{-k^2/(4s)} = \frac{k^2}{4s^2}e^{-k^2/(4s)}, \tag{136}$$

we write $\rho_\kappa(s)s^{-m} = s^{-2}\,\rho_\kappa(s)s^{2-m}$ and integrate by parts, first on $[\varepsilon, \xi^2]$ and then letting $\varepsilon \to 0^+$:

$$I_m(k) = \frac{4\rho_\kappa(\xi^2)\xi^{4-2m}}{k^2}e^{-k^2/(4\xi^2)} - \frac{4}{k^2}\int_0^{\xi^2}\frac{d}{ds}\big[\rho_\kappa(s)s^{2-m}\big]\,e^{-k^2/(4s)}\,ds. \tag{137}$$

The boundary term at $s = 0$ vanishes: by Eq. (119),

$$0 \le \rho_\kappa(s)s^{2-m}e^{-k^2/(4s)} \le \frac{2}{\sqrt{\pi}}\,s^{5/2-m}e^{-k^2/(4s)} \longrightarrow 0, \qquad s \to 0^+, \tag{138}$$

because the exponential factor decays faster than any power of s.

To bound the remaining integral we use $\rho_\kappa'(s) \le \pi^{-1/2} s^{-1/2}$ and Eq. (119). For $0 < s \le \xi^2$,

$$\begin{aligned} \left| s^2 \frac{d}{ds}\left[\rho_\kappa(s) s^{2-m}\right] \right| &= \left| s^{4-m}\rho_\kappa'(s) + (2-m) s^{3-m}\rho_\kappa(s) \right| \\ &\le \frac{1+2|m-2|}{\sqrt{\pi}}\, s^{7/2-m} \le \frac{1+2|m-2|}{\sqrt{\pi}}\, \xi^{7-2m}, \end{aligned} \tag{139}$$

where the last step uses $7/2 - m \ge 0$. Hence, with $\int_0^{\xi^2} s^{-2} e^{-k^2/(4s)}\, ds = (4/k^2) e^{-k^2/(4\xi^2)}$,

$$\left| \frac{4}{k^2} \int_0^{\xi^2} \frac{d}{ds}\left[\rho_\kappa(s) s^{2-m}\right] e^{-k^2/(4s)}\, ds \right| \le \frac{16\,(1+2|m-2|)\,\xi^{7-2m}}{\sqrt{\pi}\, k^4} e^{-k^2/(4\xi^2)}, \tag{140}$$

which is $\mathcal{O}\big(k^{-4} e^{-k^2/(4\xi^2)}\big)$. The same bound shows that the integral in Eq. (137) converges absolutely, which justifies the limit $\varepsilon \to 0^+$. Since $\rho_\kappa(\xi^2) > 0$,

$$I_m(k) = \frac{4\rho_\kappa(\xi^2)\xi^{4-2m}}{k^2} e^{-k^2/(4\xi^2)} \left[1 + \mathcal{O}(k^{-2})\right], \qquad k \to \infty. \tag{141}$$

With $m = 5/2$ and $m = 7/2$,

$$A_{\kappa,\xi}(k) = \frac{2\pi^{3/2}}{\xi k^2} \rho_\kappa(\xi^2) e^{-k^2/(4\xi^2)} \left[1 + \mathcal{O}(k^{-2})\right], \tag{142}$$

$$k^2 B_{\kappa,\xi}(k) = -\frac{\pi^{3/2}}{\xi^3} \rho_\kappa(\xi^2) e^{-k^2/(4\xi^2)} \left[1 + \mathcal{O}(k^{-2})\right]. \tag{143}$$

Thus

$$\frac{A_{\kappa,\xi}(k)}{k^2 B_{\kappa,\xi}(k)} = -\frac{2\xi^2}{k^2}\left[1 + \mathcal{O}(k^{-2})\right], \qquad \lambda_\parallel(k) = A_{\kappa,\xi}(k) + k^2 B_{\kappa,\xi}(k) = k^2 B_{\kappa,\xi}(k)\left[1 + \mathcal{O}(k^{-2})\right], \tag{144}$$

which is Eq. (14); in particular $\lambda_\parallel(k) < 0$ for all sufficiently large k. The eigenvalues of the symmetric matrix $\widehat{K}^L_\kappa(\boldsymbol{k})$ are $\lambda_\perp(k) = A_{\kappa,\xi}(k) > 0$ (twice) and $\lambda_\parallel(k)$, so its spectral norm is $\max\{\lambda_\perp(k), |\lambda_\parallel(k)|\}$. Because $\lambda_\perp(k)/|\lambda_\parallel(k)| = \mathcal{O}(k^{-2})$, the norm equals $|\lambda_\parallel(k)|$ for large k, which gives Eq. (15).

Tail quantities. Since $\|\theta_S(r) P(\boldsymbol{r})\|_2 = \theta_S(r)$, spherical coordinates give

$$E_S(r_c) = 4\pi \int_{r_c}^\infty r^2 \theta_S(r)\, dr, \qquad E_F(k_c) = \frac{1}{2\pi^2} \int_{k_c}^\infty k^2 \left\| \widehat{K}^L_\kappa(k) \right\|_2 dk. \tag{145}$$

For $c > 0$, integration by parts gives

$$\int_{x_0}^\infty x^2 e^{-cx^2}\, dx = \frac{x_0}{2c} e^{-cx_0^2} + \frac{1}{2c} \int_{x_0}^\infty e^{-cx^2}\, dx, \tag{146}$$

and the last integral is at most $e^{-cx_0^2}/(2cx_0)$. Hence

$$\int_{r_c}^\infty r^2 e^{-\xi^2 r^2}\, dr \sim \frac{r_c}{2\xi^2} e^{-\xi^2 r_c^2}, \qquad \int_{k_c}^\infty k^2 e^{-k^2/(4\xi^2)}\, dk \sim 2\xi^2 k_c e^{-k_c^2/(4\xi^2)}. \tag{147}$$

Because the integrands are asymptotically equivalent to $\rho_\kappa(\xi^2) e^{-\xi^2 r^2}$ and $(\pi^{3/2}/\xi^3)\rho_\kappa(\xi^2) e^{-k^2/(4\xi^2)}$, respectively, the tails inherit these rates:

$$E_S(r_c) \sim 4\pi \rho_\kappa(\xi^2) \frac{r_c}{2\xi^2} e^{-\xi^2 r_c^2}, \qquad E_F(k_c) \sim \frac{\rho_\kappa(\xi^2)}{2\sqrt{\pi}\,\xi^3} 2\xi^2 k_c e^{-k_c^2/(4\xi^2)}, \tag{148}$$

which is Eq. (17).

D Pair-response operator bounds

Lemma D.1 (Pair-response perturbation)**.** *Let Γ and Γ' be built by the degree-minus-weight construction, $(\Gamma X)_i = \sum_{j\neq i} K_{ij}(x_i - x_j)$, from pair responses K_{ij} and K'_{ij}, and let $E_{ij} = K'_{ij} - K_{ij}$. If $E_{ij} = E_{ji} = E_{ij}^{\mathsf{T}}$ for all $i \neq j$, then*

$$\|\Gamma' - \Gamma\|_2 \leq 2 \max_i \sum_{j\neq i} \|E_{ij}\|_2. \tag{149}$$

Proof. The difference $\Gamma' - \Gamma$ is symmetric, and pairing the (i,j) and (j,i) terms gives

$$X^{\mathsf{T}}(\Gamma' - \Gamma)X = \frac{1}{2}\sum_{i\neq j}(x_i - x_j)^{\mathsf{T}} E_{ij}(x_i - x_j). \tag{150}$$

Using $|x_i - x_j|^2 \leq 2|x_i|^2 + 2|x_j|^2$ and the symmetry in i, j,

$$\left|X^{\mathsf{T}}(\Gamma' - \Gamma)X\right| \leq 2\sum_i |x_i|^2 \sum_{j\neq i} \|E_{ij}\|_2 \leq 2\max_i \sum_{j\neq i} \|E_{ij}\|_2 \, |X|^2. \tag{151}$$

For a symmetric operator the spectral norm is the supremum of the quadratic form over unit vectors. □

The hypothesis holds for every even, pointwise symmetric pair kernel. It also holds for the particle–mesh response, because an even, real-symmetric influence function makes each block of W_h real and symmetric and $(W_h)_{ij} = (W_h)_{ji}$. Lemma D.1 with $E_{ij} = R_{k_c}(q_i - q_j)$ gives Eq. (25). Since $|e^{i\boldsymbol{k}\cdot\boldsymbol{r}}| = 1$, every term satisfies $\|R_{k_c}(\boldsymbol{r})\|_2 \leq V^{-1}\sum_{|\boldsymbol{k}|>k_c} \|\widehat{K}^L_\kappa(\boldsymbol{k})\|_2$, and the sum over $j \neq i$ has $N-1$ terms, which gives Eq. (26). Lemma D.1 with $E_{ij} = K^h_{ij} - K_{ij}$ gives Eq. (38).

Proof of Proposition 3.2. Both terms in Eq. (40) are symmetric. Since $\widetilde{\Gamma}_h$ has zero block-row sums, it annihilates $\mathcal{T}$, and by symmetry it maps $\operatorname{Range}\Pi = \mathcal{T}^\perp$ into itself. Hence $\Gamma_h t = \widetilde{\Gamma}_h t + \delta_h \Pi t = 0$ for $t \in \mathcal{T}$. For arbitrary X, write $Y = \Pi X$. Then

$$X^{\mathsf{T}}\Gamma_h X = Y^{\mathsf{T}}\widetilde{\Gamma}_h Y + \delta_h |Y|^2 \geq \left(\lambda_{\min}\big(\widetilde{\Gamma}_h|_{\operatorname{Range}\Pi}\big) + \delta_h\right)|Y|^2 \geq 0 \tag{152}$$

by the choice of δ_h. Finally, $\|\Gamma_h - \Gamma\|_2 \leq \|\widetilde{\Gamma}_h - \Gamma\|_2 + \delta_h\|\Pi\|_2 \leq \epsilon_{\mathrm{op}} + \delta_h$ by Eq. (38) and $\|\Pi\|_2 = 1$. □

E Particle–mesh alias response

Single-mode pair response. Spreading a particle at q onto the mesh and taking the discrete Fourier transform produces, for a retained mode $\boldsymbol{k}$, the deposited amplitude $S_h(\boldsymbol{k}, q)$: sampling the window on the mesh folds all aliases $\boldsymbol{k}_n$ onto $\boldsymbol{k}$. Because the assignment window is real and even, $\widehat{W}_h$ is real, and gathering the field of mode $\boldsymbol{k}$ at q_i produces the factor $\overline{S_h(\boldsymbol{k}, q_i)}$. Hence the contribution of mode $\boldsymbol{k}$ to $(W_h)_{ij}$ is proportional to

$$\overline{S_h(\boldsymbol{k}, q_i)}\, G_h(\boldsymbol{k})\, S_h(\boldsymbol{k}, q_j) = H_h(\boldsymbol{k})\, e^{i\boldsymbol{k}\cdot(q_i - q_j)}\,(1 + \alpha_i^*)\,(1 + \alpha_j)\,, \tag{153}$$

with the same normalization as the exact contribution $\widehat{K}^L_\kappa(\boldsymbol{k})e^{i\boldsymbol{k}\cdot(q_i-q_j)}$. Subtracting the exact contribution and expanding the product gives Eq. (51).

B-spline alias ratio. Write $x_\ell = k_\ell h/2$. The ℓ-th component of $\boldsymbol{k}_n$ satisfies $(\boldsymbol{k}_n)_\ell h/2 = x_\ell + \pi n_\ell$, and $\sin(x_\ell + \pi n_\ell) = (-1)^{n_\ell} \sin x_\ell$. Therefore

$$\left| \frac{\widehat{W}_h(\boldsymbol{k}_n)}{\widehat{W}_h(\boldsymbol{k})} \right| = \prod_{\ell : n_\ell \neq 0} \left| \frac{k_\ell h/2}{k_\ell h/2 + \pi n_\ell} \right|^p . \tag{154}$$

For a mode resolved by the mesh, $|x_\ell| \le \pi/2$, so $|x_\ell + \pi n_\ell| \ge \pi |n_\ell|/2$ and each factor is at most $(h|k_\ell|/\pi)^p |n_\ell|^{-p}$, where $h|k_\ell|/\pi \le 1$ and $|k_\ell| \le |\boldsymbol{k}|$. Every $n \neq 0$ has at least one nonzero component, so the product of the factors $(h|k_\ell|/\pi)^p$ is at most $(h|\boldsymbol{k}|/\pi)^p$ and

$$|\alpha_h(\boldsymbol{k}, q)| \le \left(\frac{h|\boldsymbol{k}|}{\pi} \right)^p \sum_{n \neq 0} \prod_{\ell : n_\ell \neq 0} |n_\ell|^{-p} = \left[\left(1 + 2 \sum_{m=1}^{\infty} m^{-p} \right)^3 - 1 \right] \left(\frac{h|\boldsymbol{k}|}{\pi} \right)^p , \tag{155}$$

uniformly in q. The series converges for $p \ge 2$, which gives $\alpha_h(\boldsymbol{k}, q) = \mathcal{O}((h|\boldsymbol{k}|)^p)$.

Mesh law. For ideal deconvolution, $D_h = 0$, and the bound on α_h gives, for each pair,

$$\|E_{ij}^h\|_2 \lesssim \frac{1}{V} \sum_{|\boldsymbol{k}| \le k_c} \|\widehat{K}_\kappa^L(\boldsymbol{k})\|_2 \left[(h|\boldsymbol{k}|)^p + (h|\boldsymbol{k}|)^{2p} \right] , \tag{156}$$

with a constant depending only on p. Summing over $j \neq i$ multiplies the bound by $N - 1$. The lattice sum $V^{-1} \sum_{\boldsymbol{k}} |\boldsymbol{k}|^p \|\widehat{K}_\kappa^L(\boldsymbol{k})\|_2$ is a Riemann sum for $M_p(\kappa, \xi)$ and is finite by the Gaussian decay in Eq. (15). Since $h|\boldsymbol{k}| \le \sqrt{3}\,\pi$ on the resolved modes, the h^{2p} term is subdominant, which gives Eq. (58) for a fixed configuration and cutoff.

F Finite-time thermostat

Exact covariance. For frozen positions, consider the Ornstein–Uhlenbeck dynamics

$$dp = -\frac{1}{m} \Gamma_h p \, dt + \left(2\beta^{-1} \Gamma_h \right)^{1/2} dW_t, \tag{157}$$

where W_t is a standard Wiener process. Over a step of length Δt, the solution is $p(\Delta t) = R_{\Delta t}(\Gamma_h) p(0) + \eta$ with a Gaussian increment η of mean zero. By the Itô isometry and the spectral calculus,

$$\mathrm{Cov}(\eta) = \int_0^{\Delta t} e^{-2\tau \Gamma_h / m} \, 2\beta^{-1} \Gamma_h \, d\tau = \beta^{-1} m \left(I - R_{\Delta t}(\Gamma_h)^2 \right) = S_{\Delta t}(\Gamma_h)^2, \tag{158}$$

where both sides vanish on the null space of Γ_h. Since $f_{\Delta t}(\lambda)^2 = \beta^{-1} m (1 - e^{-2\Delta t \lambda / m})$, we have $S_{\Delta t}(\Gamma_h) = f_{\Delta t}(\Gamma_h)$, and the update $p \mapsto R_{\Delta t}(\Gamma_h) p + f_{\Delta t}(\Gamma_h) \xi$ reproduces the exact frozen-configuration transition law.

Momentum conservation. Let $t \in \mathcal{T}$. Since $\Gamma_h t = 0$, $R_{\Delta t}(\Gamma_h) t = t$ and $f_{\Delta t}(\Gamma_h) t = f_{\Delta t}(0) t = 0$. Both operators are symmetric, so

$$t^{\mathsf{T}} p^+ = \left(R_{\Delta t}(\Gamma_h) t \right)^{\mathsf{T}} p + \left(f_{\Delta t}(\Gamma_h) t \right)^{\mathsf{T}} \xi = t^{\mathsf{T}} p \tag{159}$$

for every noise sample. The Lanczos approximation retains this property because the starting vector $z = \Pi \xi$ and the reconstructed increment are projected onto $\mathrm{Range}\, \Pi = \mathcal{T}^\perp$, and the center-of-mass component is copied exactly through the damping step.

Invariance of the conditional Maxwell law. Suppose p is Gaussian with fixed total momentum and covariance $\beta^{-1}m\Pi$. Since $R_{\Delta t}(\Gamma_h)$ commutes with Π and equals the identity on $\mathcal{T}$, $I - R_{\Delta t}(\Gamma_h)^2 = \Pi - R_{\Delta t}(\Gamma_h)^2\Pi$. Hence

$$\mathrm{Cov}(p^+) = \beta^{-1}m\left(R_{\Delta t}(\Gamma_h)^2\Pi + \Pi - R_{\Delta t}(\Gamma_h)^2\Pi\right) = \beta^{-1}m\Pi, \tag{160}$$

and the total momentum is unchanged. The conditional Maxwell law is therefore invariant under the thermostat step.

Range-start function. Expanding the exponential gives $f_{\Delta t}(\lambda)^2 = 2\beta^{-1}\Delta t\,\lambda + \mathcal{O}(\lambda^2)$ as $\lambda \to 0^+$. Hence the constant in the range-start analysis is $C = (2\beta^{-1}\Delta t)^{1/2}$, and $g(\lambda) = f_{\Delta t}(\lambda)/\lambda$ has the stated $\lambda^{-1/2}$ singularity.

G Equal-accuracy timing of the splitting parameter

This appendix keeps the equal-accuracy timing study that informs the parameter choice of Section 5. Every run was required to meet the same measured relative operator tolerance on the two dense reference configurations A ($N = 30$, $L = 6$) and B ($N = 40$, $L = 10$), $\kappa = 1$; the cost per graph action was then timed on an $N = 4000$ configuration (Table 8, Fig. 16). All 44 accepted runs met the tolerance on both configurations and were positive semidefinite and certified. The required Ewald scale was only 1.0–1.075 $\times \sqrt{\log(2/\mathrm{tol})}$, and for the tested density, hardware and tolerances the lowest cost occurred near $\xi = 0.5$–0.7. This is an empirical work-balance result, not a universal optimum. The accuracy is established on cases A and B only; it is not a guarantee for other configurations.

Tolerance	$\xi = 0.5$	0.7	1.0	1.4	2.0
10^{-5}	164	129	270	663	1999
10^{-7}	332	311	532	1366	8145
10^{-9}	592	739	2791	12704	—

Table 8: Milliseconds per graph action at equal operator accuracy (accuracy measured on cases A and B, timing at $N = 4000$).

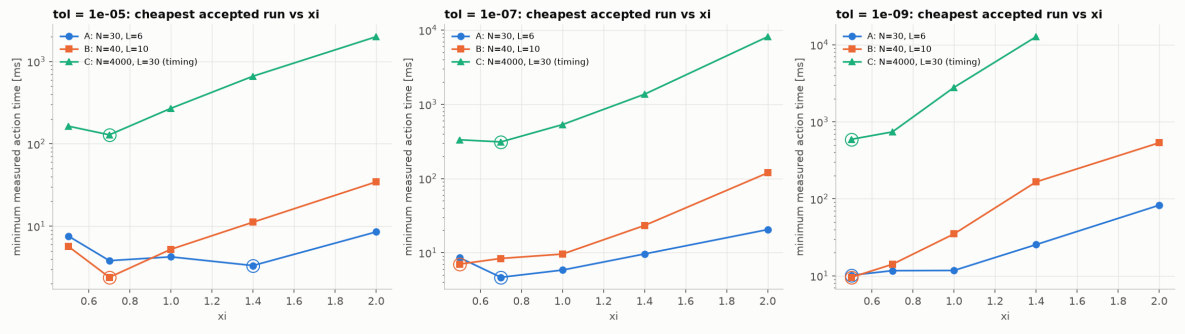


Figure 16: Cheapest accepted run (minimum measured time per graph action) versus ξ at tolerances 10^{-5}, 10^{-7}, 10^{-9}, for configurations A and B and the $N = 4000$ timing configuration ($L = 30$). Source: `equal_accuracy_xi_results/min_cost_vs_xi.png`.

H Supplementary figures for Section 7

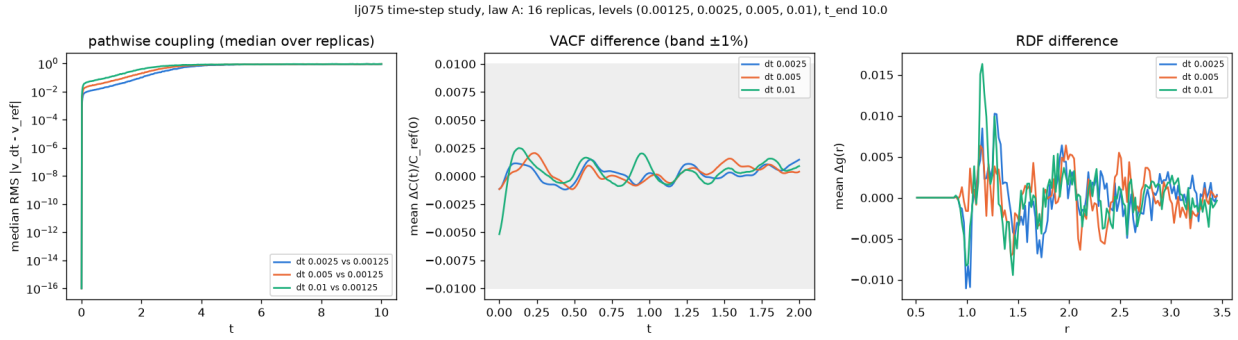


Figure 17: Time-step study, 16 coupled replicas. Left: median RMS velocity difference to the $\Delta t = 0.00125$ path. Middle and right: mean velocity-autocorrelation and radial-distribution differences (means only; verdicts use Table 7).

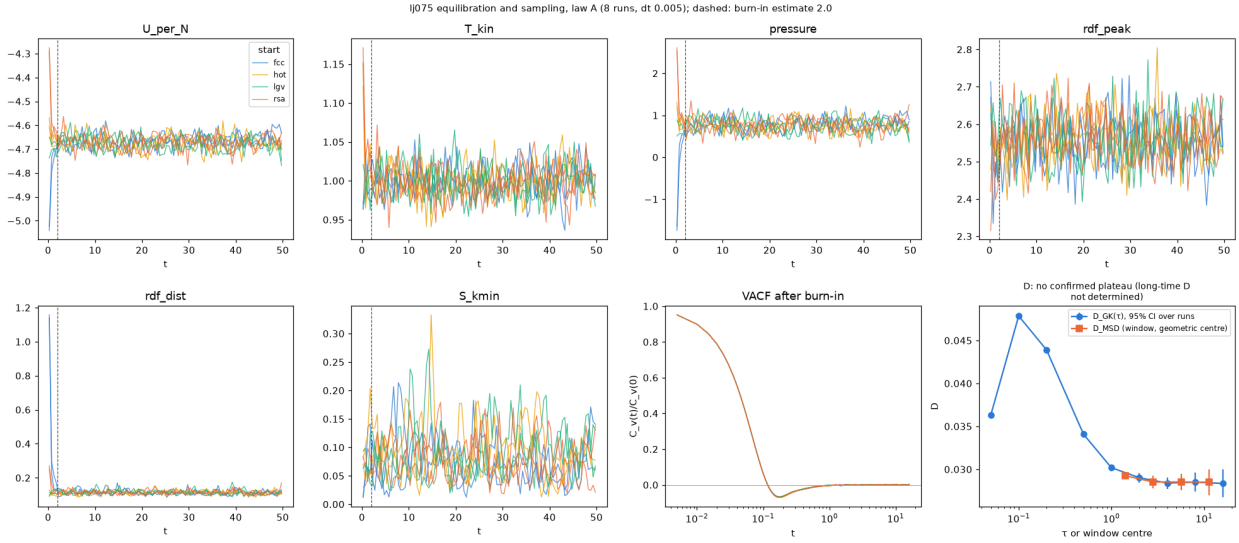


Figure 18: Equilibration: 8 runs of 50 time units at $\Delta t = 0.005$ (ranks 12/5), four start types; dashed: measured burn-in 2. Bottom right: Green–Kubo and windowed mean-square-displacement diffusion coefficients with 95% intervals over runs; no plateau is confirmed.

Local run of the cluster scripts. The cluster workflow was exercised locally with bash on the development machine, without a scheduler; it is not a cluster result. One canonical state was integrated to $t = 1$ with $\Delta t = 0.005$ and 0.01, once with PPPM and Lanczos 14/5 and once with the dense full-periodic matrix and its eigendecomposition, sharing the noise at each Δt. Restart, locking, checkpoint and output checks passed. The RMS position and velocity differences at $t = 1$ were 1.96×10^{-8} and 1.28×10^{-7} ($\Delta t = 0.005$), 2.22×10^{-8} and 1.35×10^{-7} ($\Delta t = 0.01$), and the total-momentum change stayed below 1.3×10^{-13}. A single path pair supports no statistical statement.

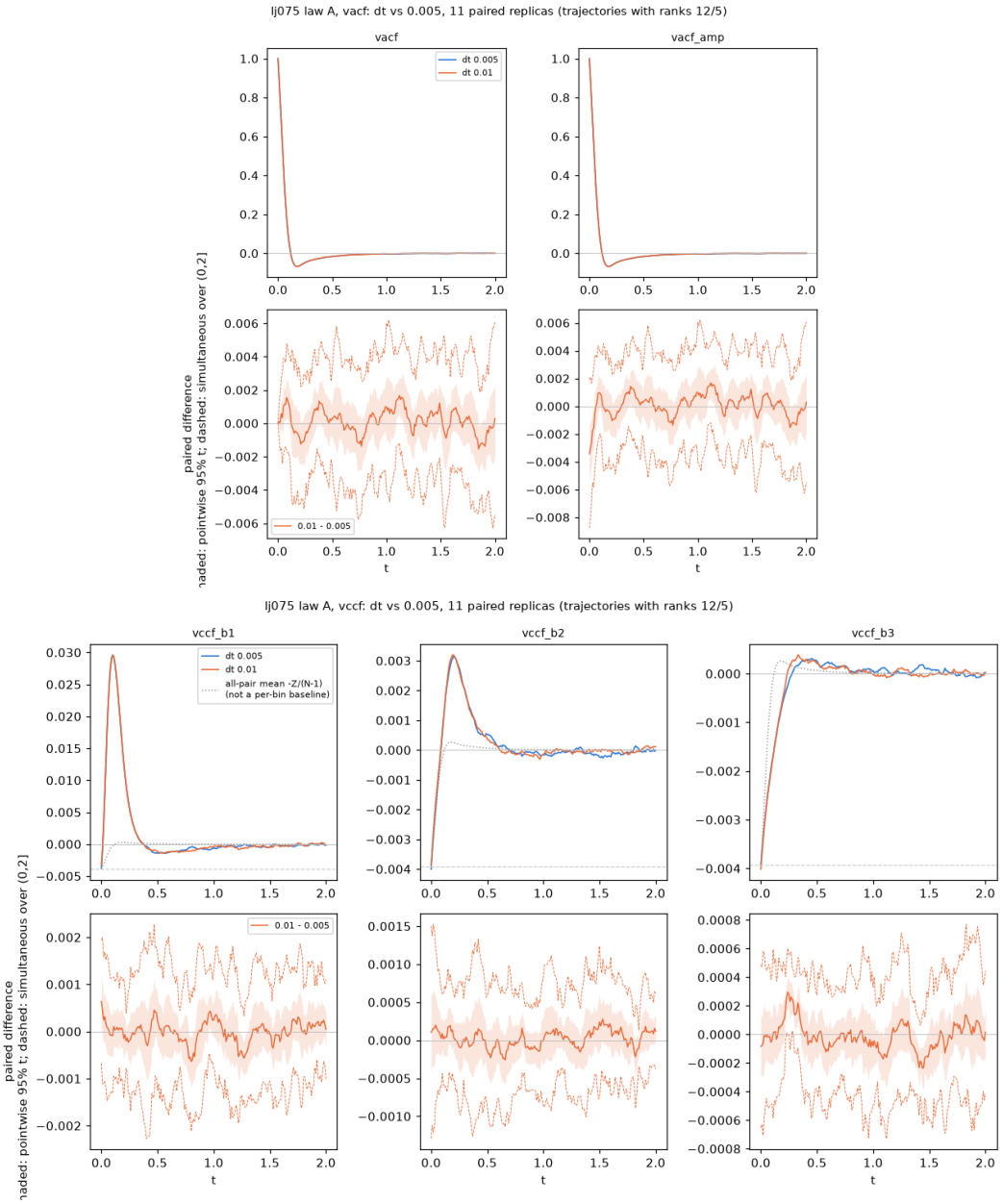


Figure 19: Second group: 11 coupled chains, $\Delta t = 0.01$ versus 0.005 only, origins $t_0 \in [10, 98]$, ranks 12/5; conventions as in Fig. 14. This group holds five of the seven FAIL cells and, lacking a finer level, cannot support display by itself.

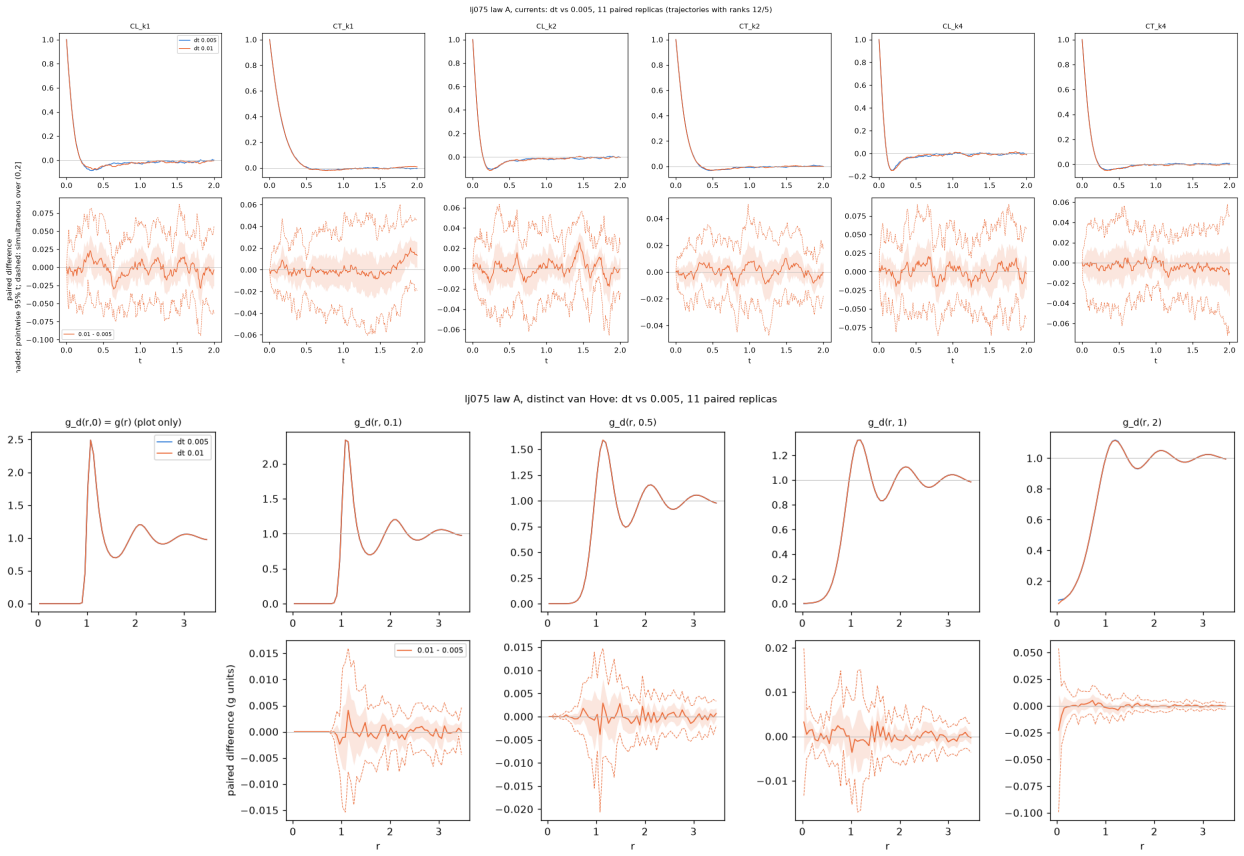


Figure 20: Second group, current correlations and $g_d(r,t)$; conventions as in Figs. 15 and 19.

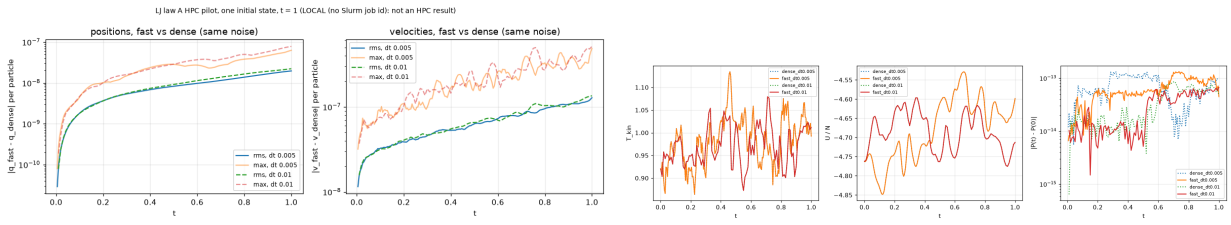


Figure 21: Local run of the cluster scripts (one state, one paired noise path per Δt, $t \le 1$). Left: PPPM–Lanczos minus dense differences of positions and velocities. Right: kinetic temperature, U/N and total momentum change for the four cases.